DIODE CIRCUITS

SEMICONDUCTOR DIODES

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

DC Input

AC Input

- Half-Wave Rectifier
- Full-Wave Center-Tapped Rectifier
- Full-Wave Bridge-Type Rectifier

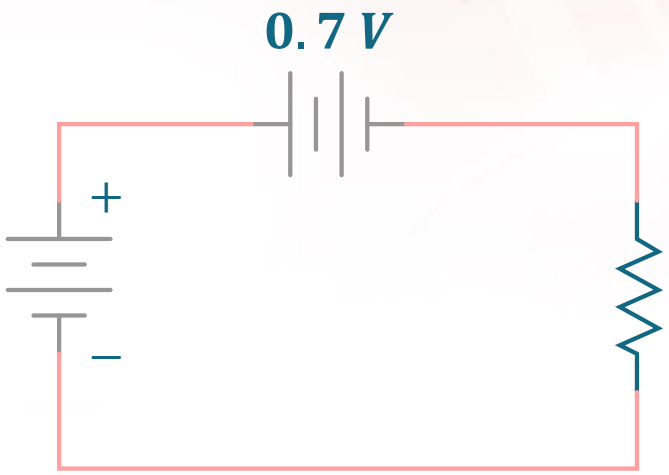
DC INPUT

DIODE CIRCUITS

DIODE STATES

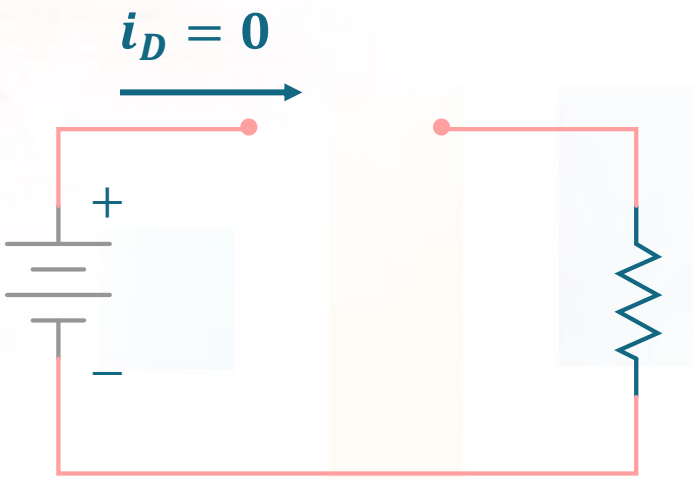
Diode is **ON**

Forward Bias Equivalent Circuit



Diode is **OFF**

Reverse Bias Equivalent Circuit



Determine the load voltage, v_L and load current, i_L for the given network.

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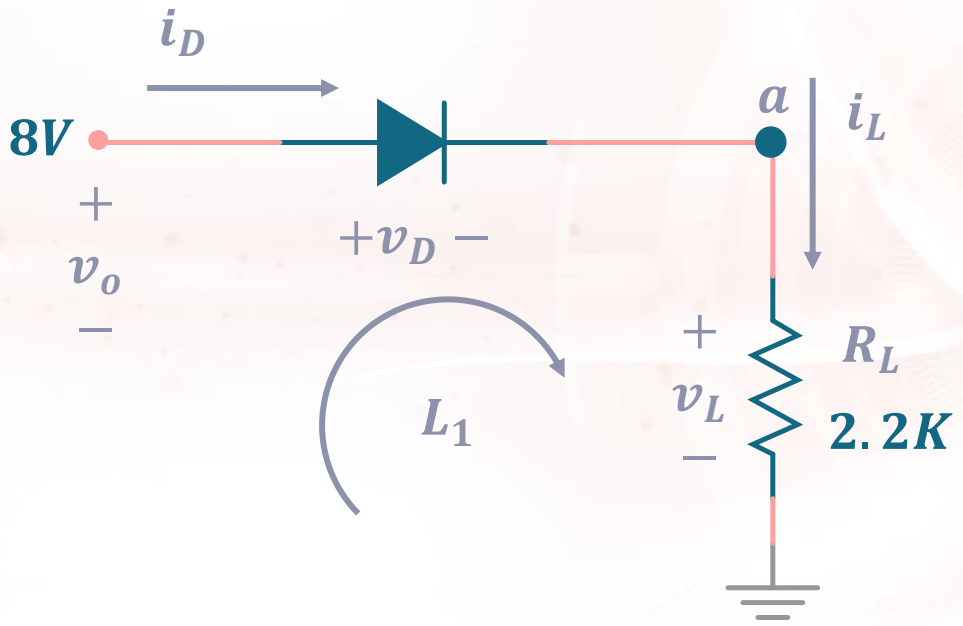


KVL @L_1,

$$-v_o + v_D + v_L = 0$$

$$v_L = v_o - v_D$$

$$v_L = 8 - 0.7$$

$$\boxed{v_L = 7.3\ V}$$

ans

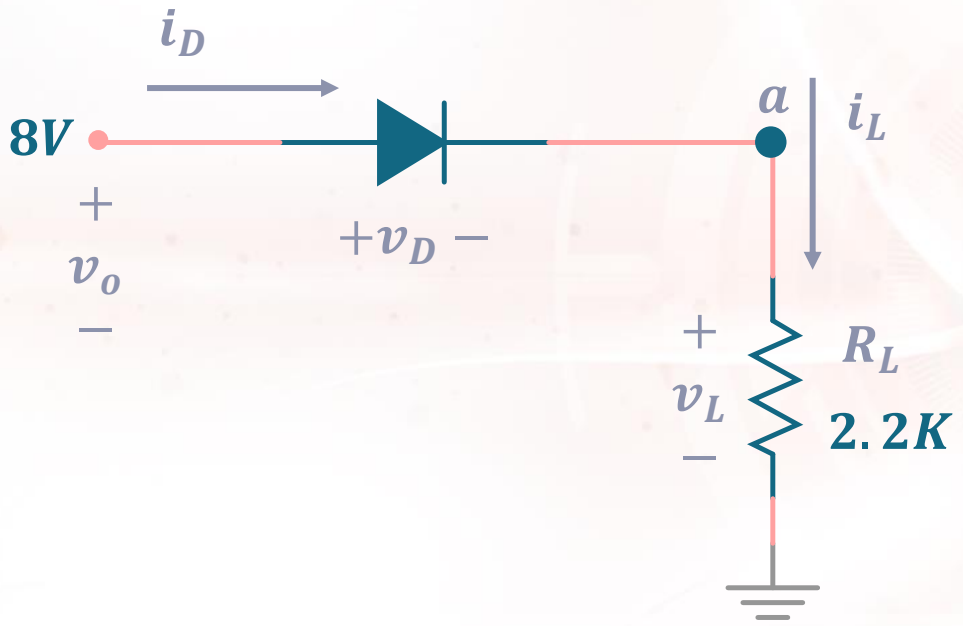
$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{7.3}{2.2K}$$

$$\boxed{i_L = 3.32\ mA}$$

ans

Determine the load voltage, v_L and load current, i_L for the given network.



Node Analysis Method,

$$v_D = v_o - v_a$$

$$v_a = v_o - v_D$$

$$v_a = 8 - 0.7$$

$$\boxed{v_L = 7.3\ V}$$

ans

$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{7.3}{2.2K}$$

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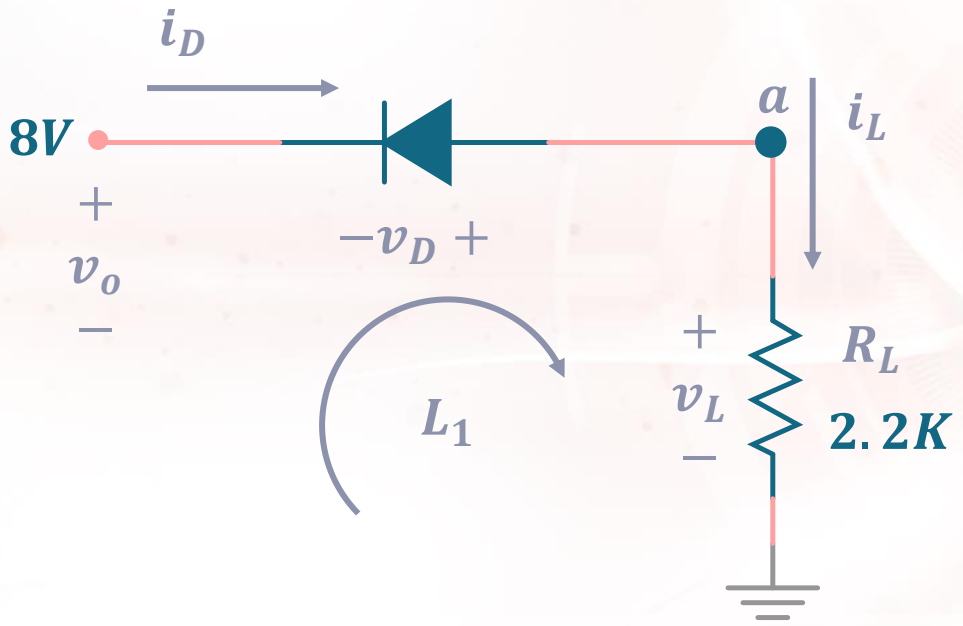

KCL @ a,

$$-i_D + i_L = 0$$

Diode is reverse bias, $i_D = 0$

$$\boxed{i_L = 0}$$

ans

$$v_L = i_L R_L$$

$$v_L = 0(R_L)$$

$$\boxed{v_L = 0}$$

ans

Determine the load voltage, v_L and load current, i_L for the given network.

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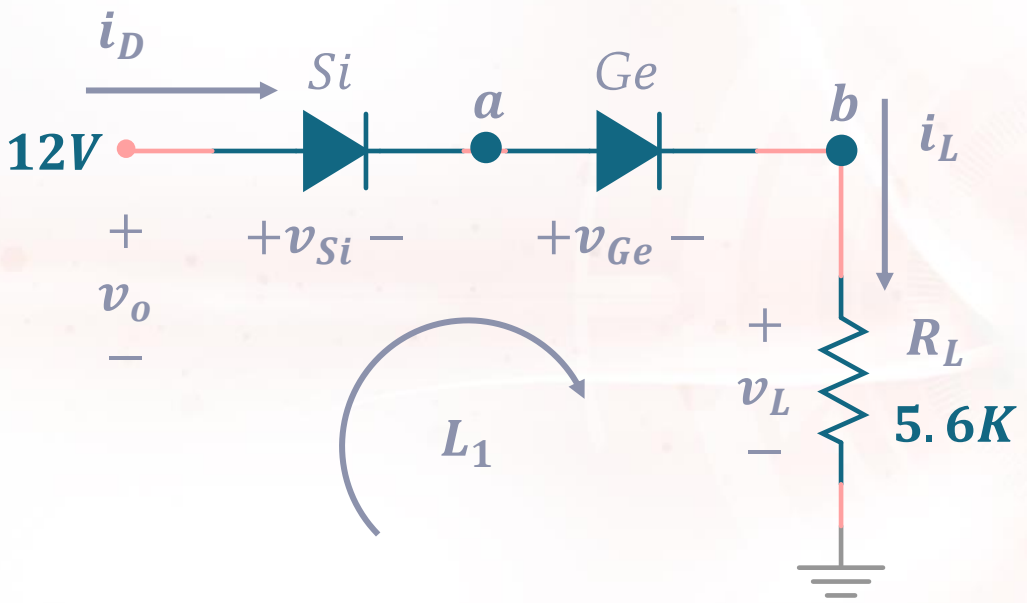


KVL @L_1,

$$-v_o + v_{Si} + v_{Ge} + v_L = 0$$

$$v_L = v_o - v_{Si} - v_{Ge}$$

$$v_L = 12 - 0.7 - 0.3$$

$$\boxed{v_L = 11\ V}$$

ans

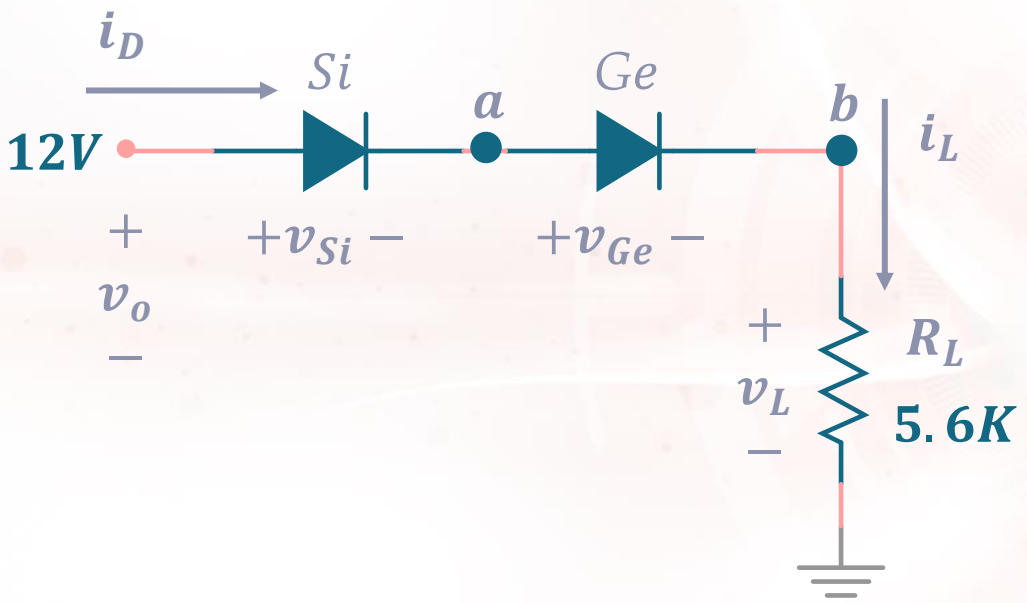
$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{11}{5.6\ K}$$

$$\boxed{i_L = 1.96\ mA}$$

ans

Determine the load voltage, v_L and load current, i_L for the given network.



Node Analysis Method ,

$$v_{Si} = v_o - v_a$$

$$v_a = v_o - v_{Si}$$

$$v_a = 12 - 0.7$$

$$v_a = 11.3\ V$$

$$v_{Ge} = v_a - v_b$$

$$v_b = v_a - v_{Ge}$$

$$v_b = 11.3 - 0.3$$

$$\boxed{v_L = 11\ V}$$

ans

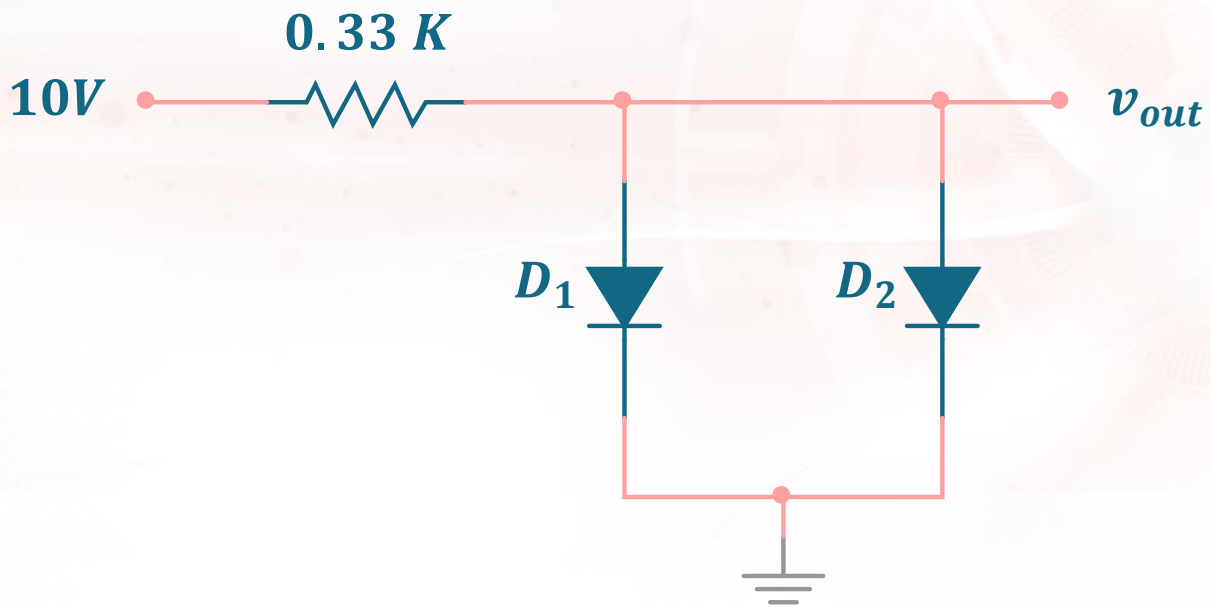
$$i_L = \frac{v_L}{R_L}$$

$$i_L = \frac{11}{5.6\ K}$$

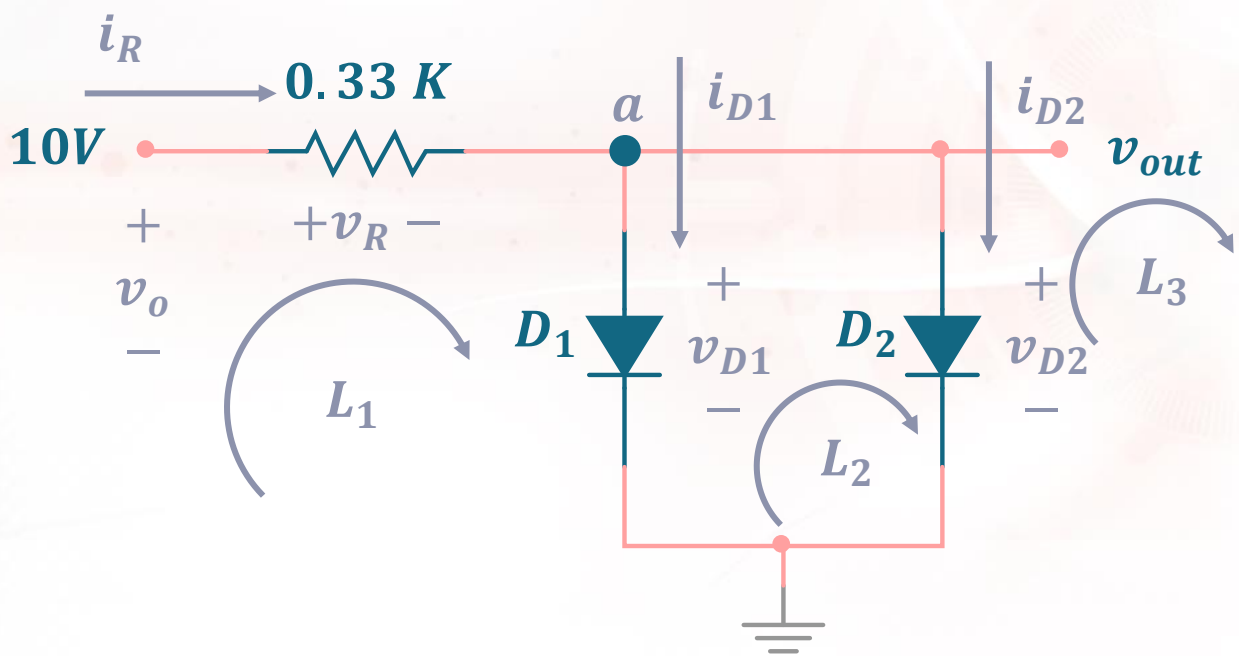
$$\boxed{i_L = 1.96\ mA}$$

ans

Determine the output voltage v_{out}, resistor current i_R, and diodes current i_{D1} and i_{D2} respectively for the given network.



Determine the output voltage v_{out}, resistor current i_R, and diodes current i_{D1} and i_{D2} respectively for the given network.



KVL @L_1,

$$-v_o + v_R + v_{D1} = 0$$

$$v_R = v_o - v_{D1}$$

$$v_R = 10 - 0.7$$

$$v_R = 9.3\ V$$

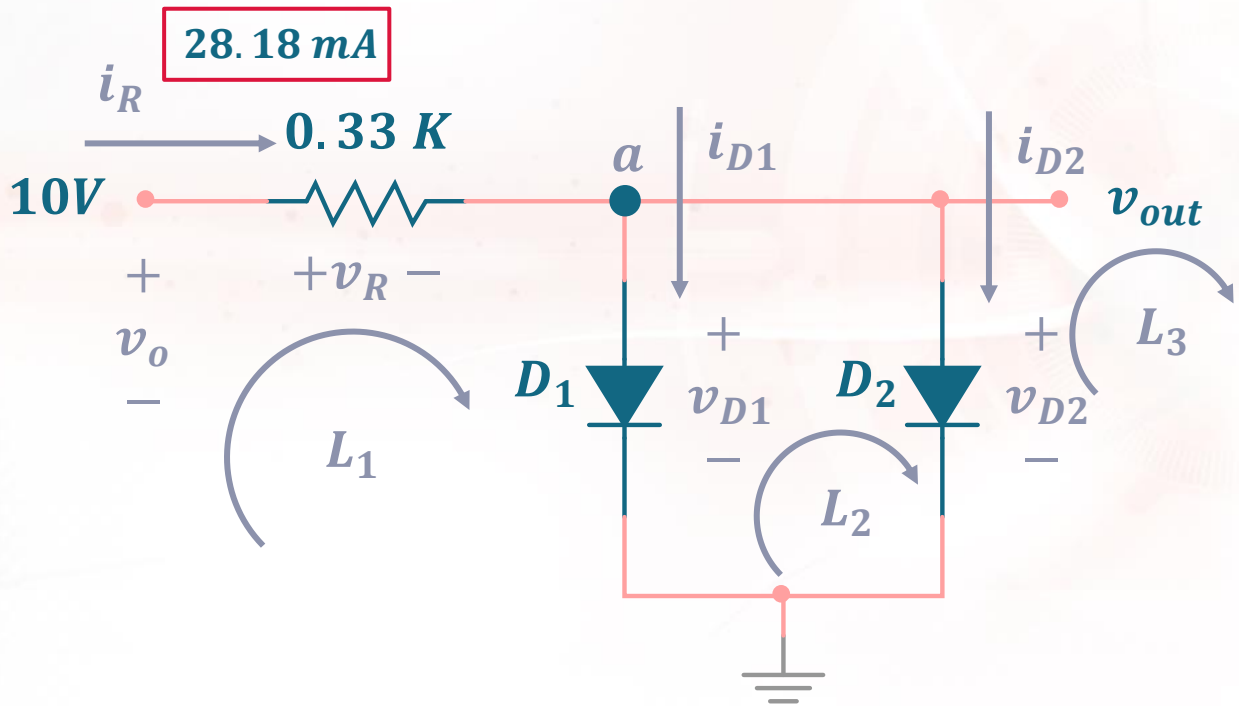
$$i_R = \frac{v_R}{R}$$

$$i_R = \frac{9.3}{0.33\ K}$$

$$\boxed{i_R = 28.18\ mA}$$

ans

Determine the output voltage v_{out}, resistor current i_R, and diodes current i_{D1} and i_{D2} respectively for the given network.



KCL @ a,

$$-i_R + i_{D1} + i_{D2} = 0$$

$$i_{D1} + i_{D2} = i_R$$

Let $i_{D1} = i_{D2}$,

$$i_{D1} + i_{D1} = i_R$$

$$2i_{D1} = i_R$$

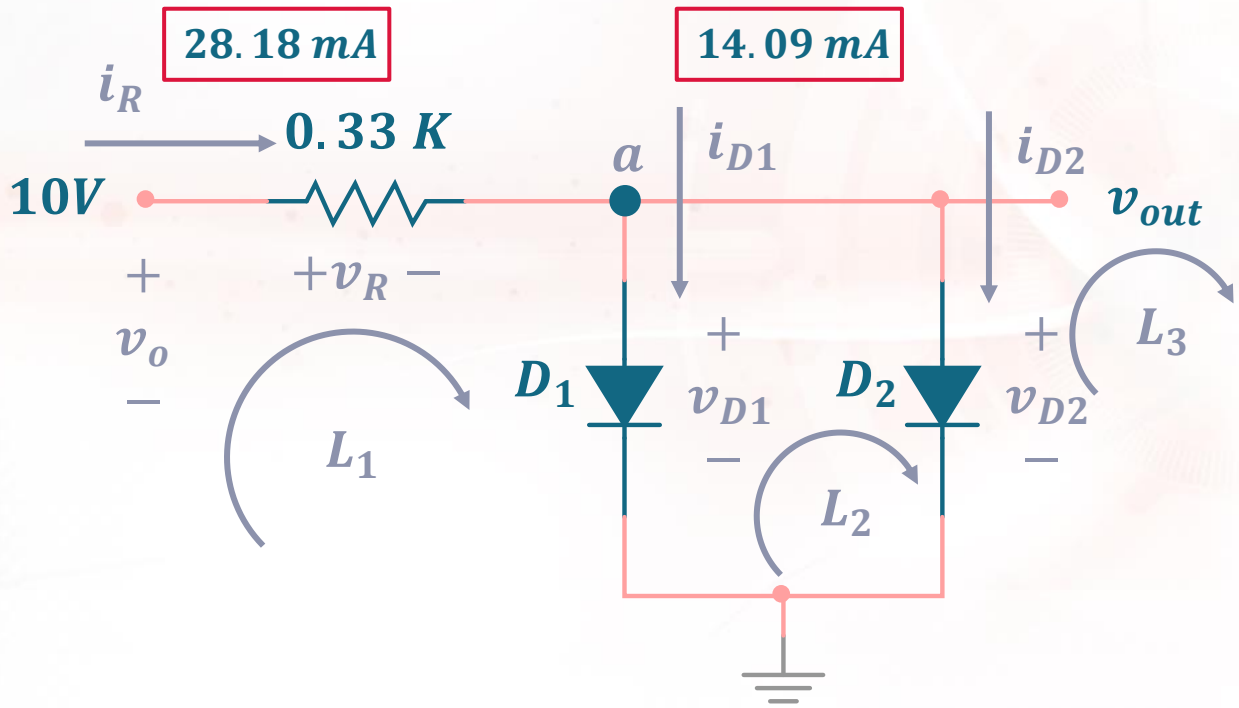
$$i_{D1} = \frac{i_R}{2}$$

$$i_{D1} = \frac{28.18\ mA}{2}$$

$$\boxed{i_{D1} = i_{D2} = 14.09\ mA}$$

ans

Determine the output voltage v_{out}, resistor current i_R, and diodes current i_{D1} and i_{D2} respectively for the given network.



KVL @ L_3,

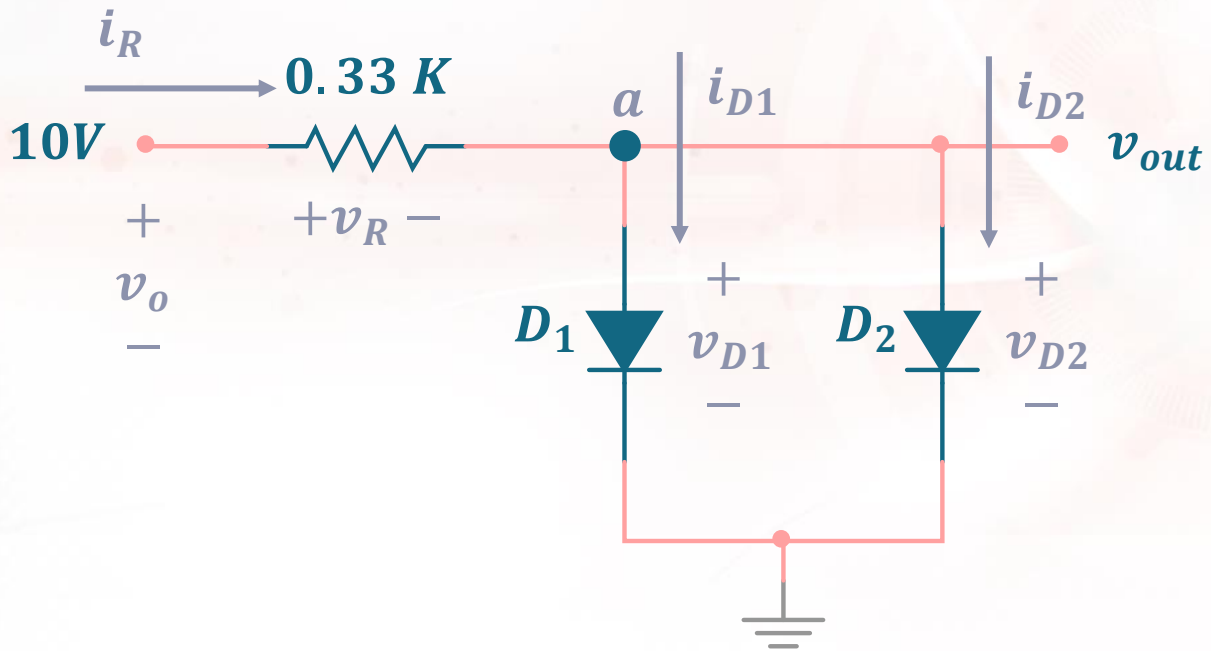
$$-v_{D2} + v_{out} = 0$$

$$v_{out} = v_{D2}$$

$$\boxed{v_{out} = 0.7\ V}$$

ans

Determine the output voltage v_{out}, resistor current i_R, and diodes current i_{D1} and i_{D2} respectively for the given network.



Node Analysis Method,

$$v_R = v_o - v_1$$

$$v_R = 10 - 0.7$$

$$v_R = 9.3\ V$$

$$i_R = \frac{v_R}{R}$$

$$i_R = \frac{9.3}{0.33\ K}$$

$$\boxed{i_R = 28.18\ mA}$$

ans

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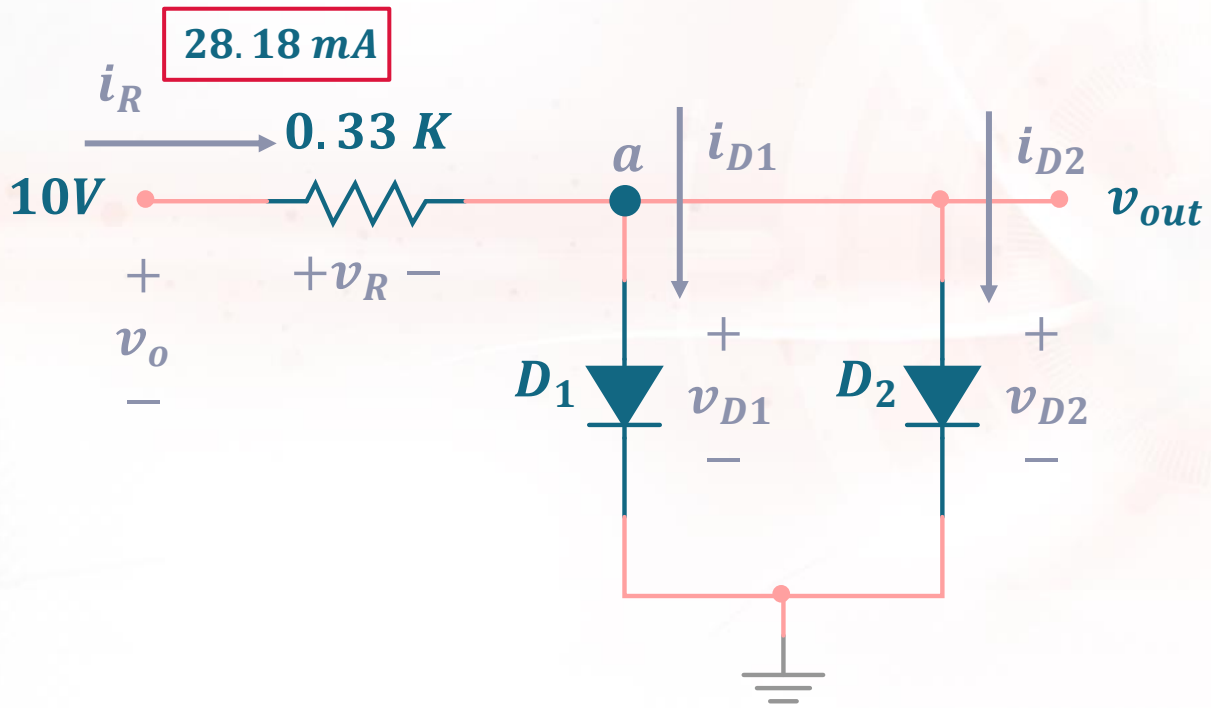

KCL @ a,

$$-i_R + i_{D1} + i_{D2} = 0$$

$$i_{D1} + i_{D2} = i_R$$

Let $i_{D1} = i_{D2}$,

$$i_{D1} + i_{D1} = i_R$$

$$2i_{D1} = i_R$$

$$i_{D1} = \frac{i_R}{2}$$

$$i_{D1} = \frac{28.18\ mA}{2}$$

$$\boxed{i_{D1} = i_{D2} = 14.09\ mA}$$

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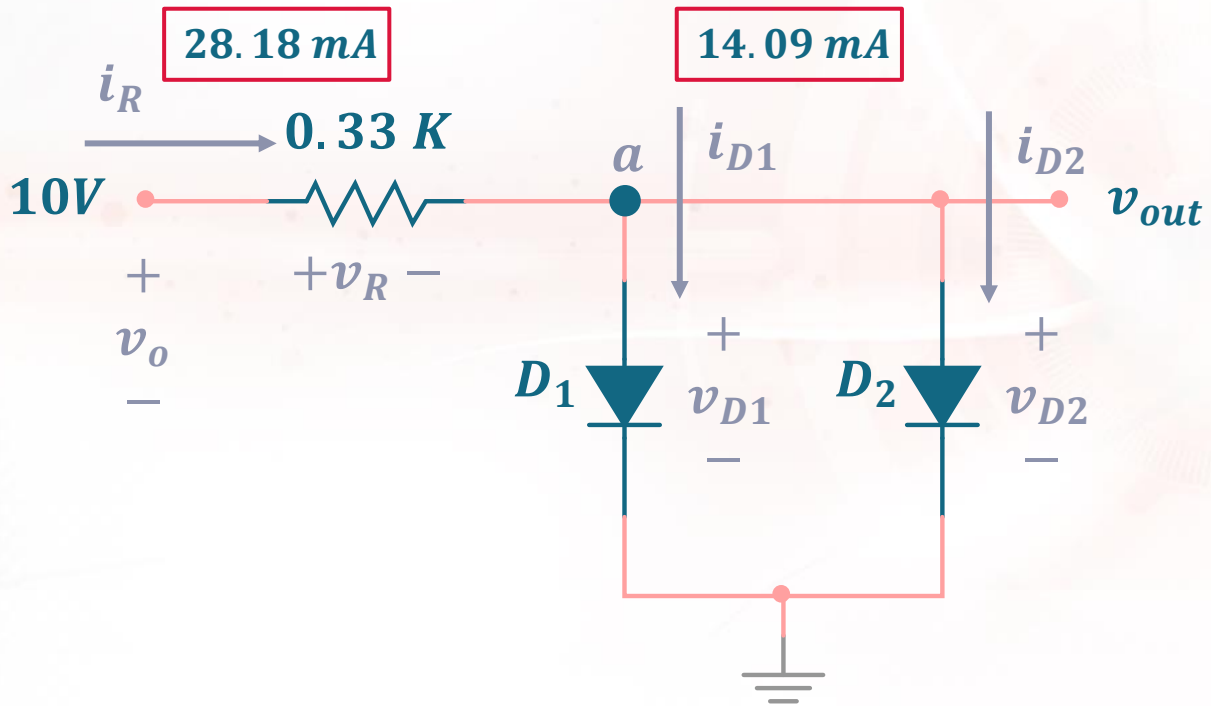


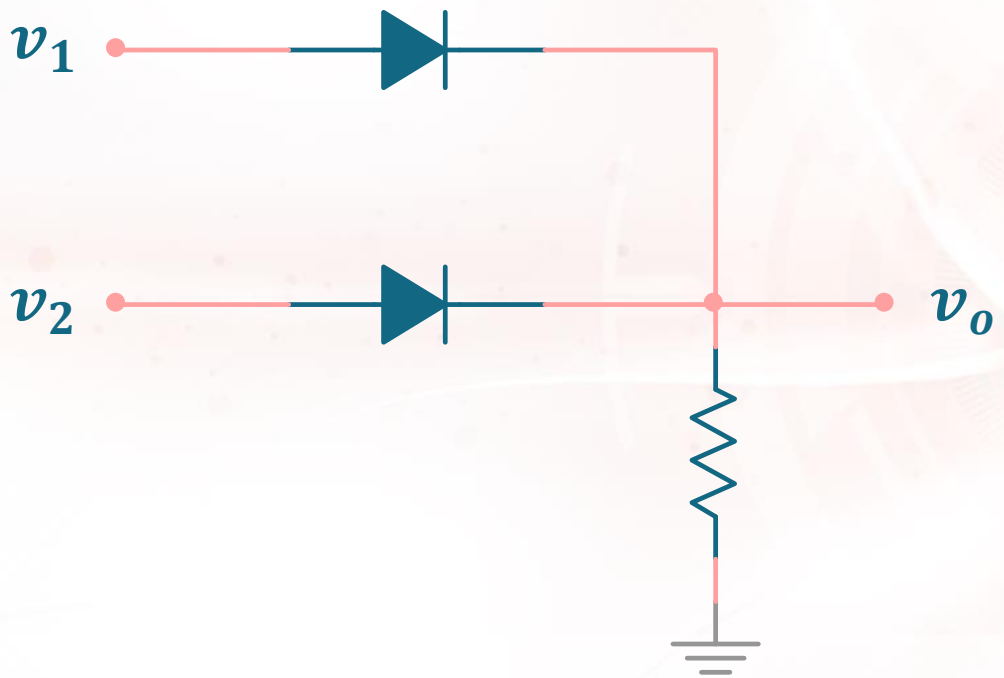
Node Analysis Method,

$$v_{D2} = v_{out} - 0$$

$$\boxed{v_{out} = 0.7\ V}$$

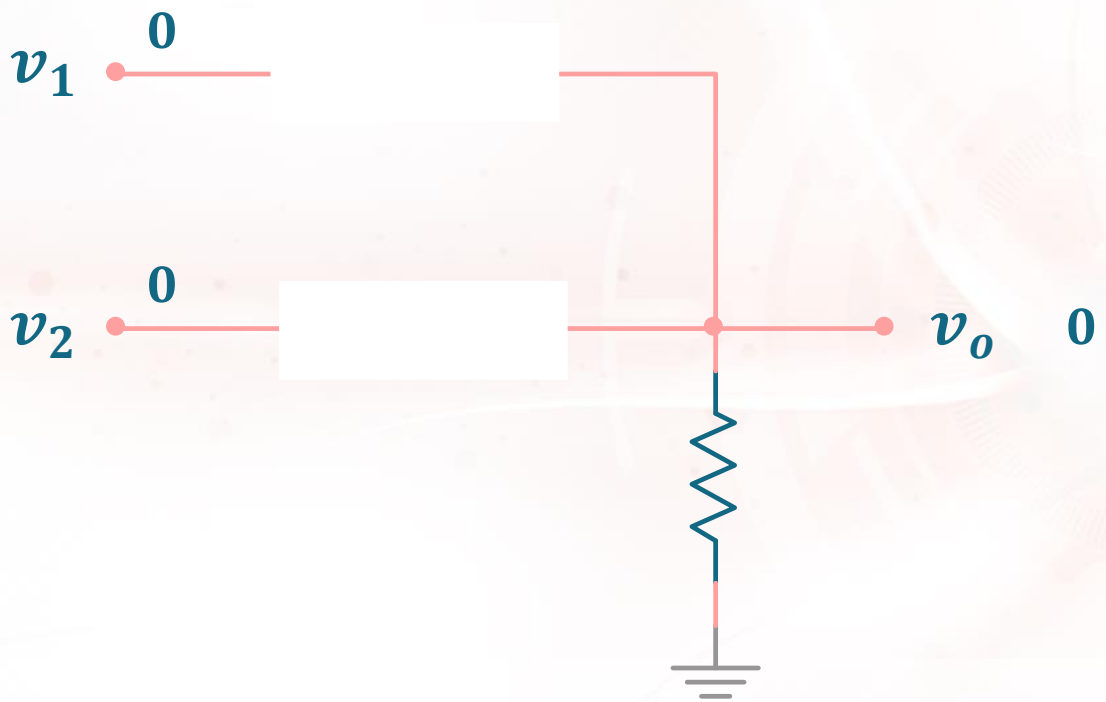
ans

POSITIVE LOGIC OR GATE



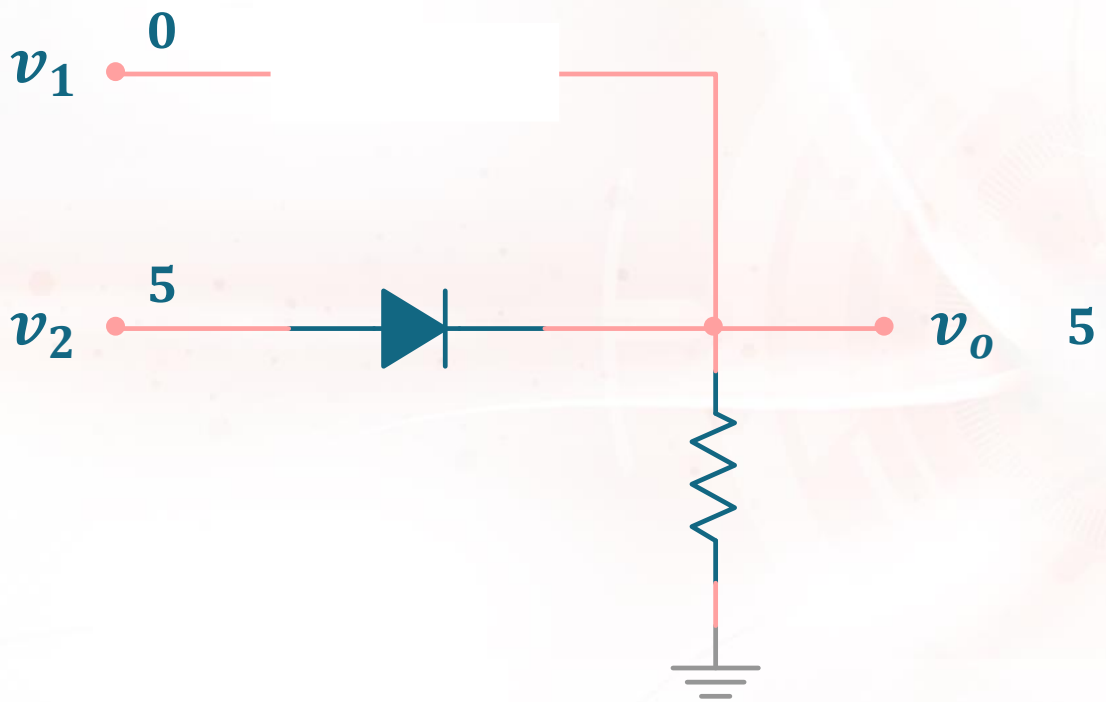
v_1	v_2	v_o
0	0	
0	5	
5	0	
5	5	

POSITIVE LOGIC OR GATE



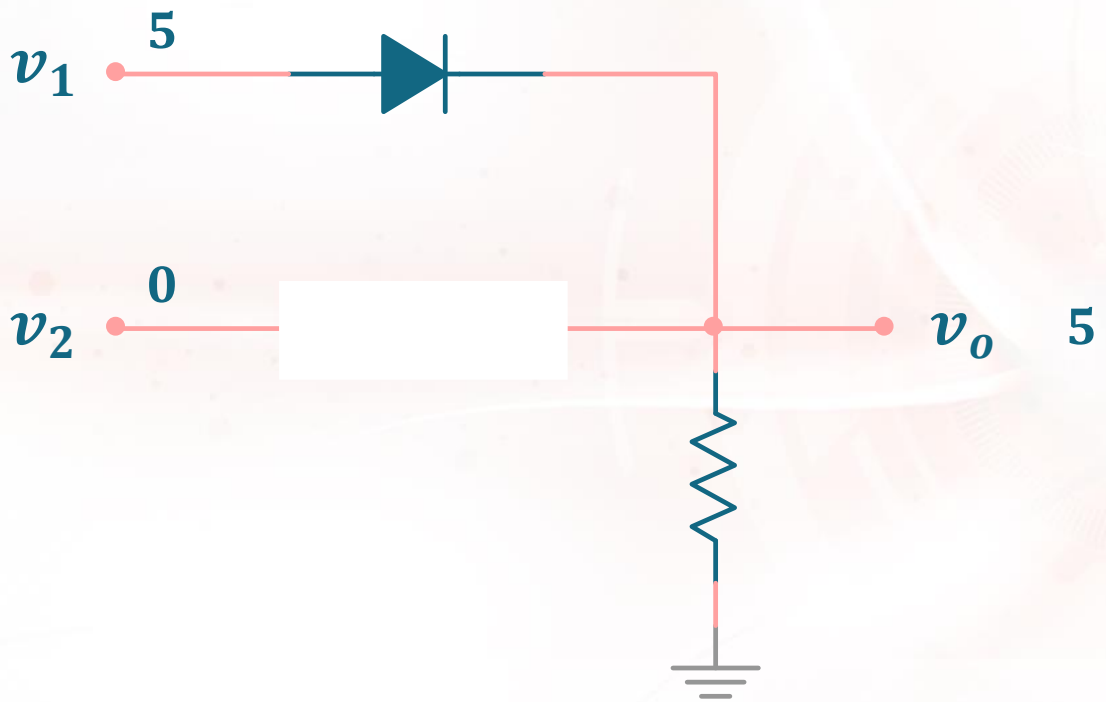
v_1	v_2	v_o
0	0	0
0	5	
5	0	
5	5	

POSITIVE LOGIC OR GATE



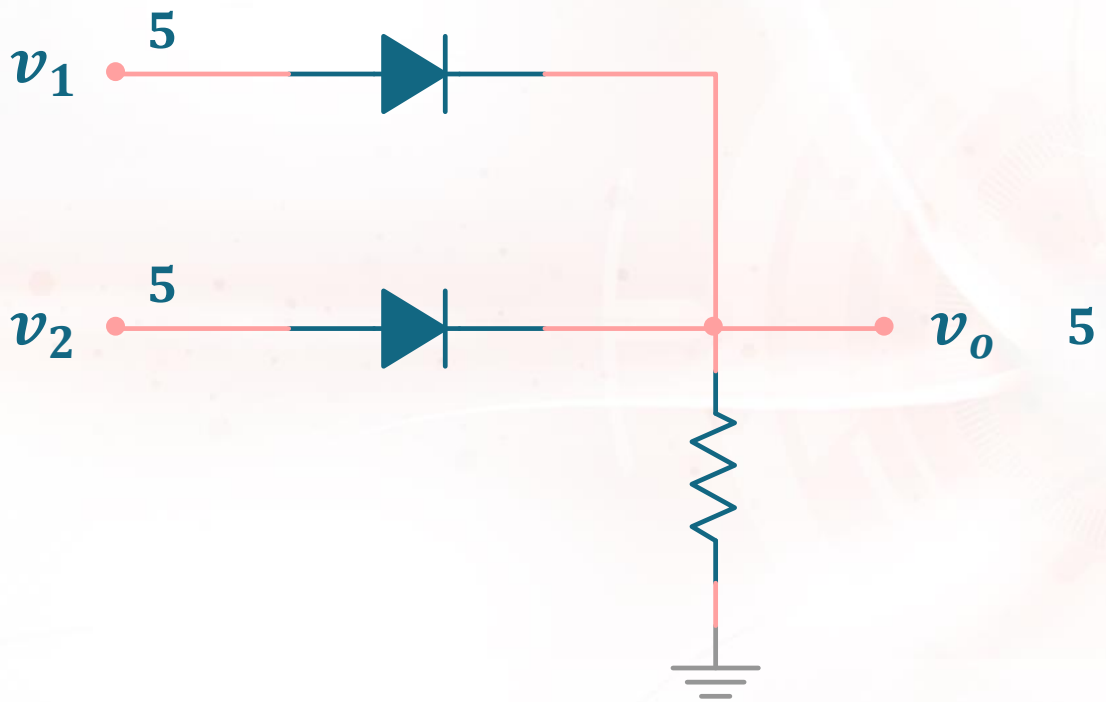
v_1	v_2	v_o
0	0	0
0	5	5
5	0	
5	5	

POSITIVE LOGIC OR GATE



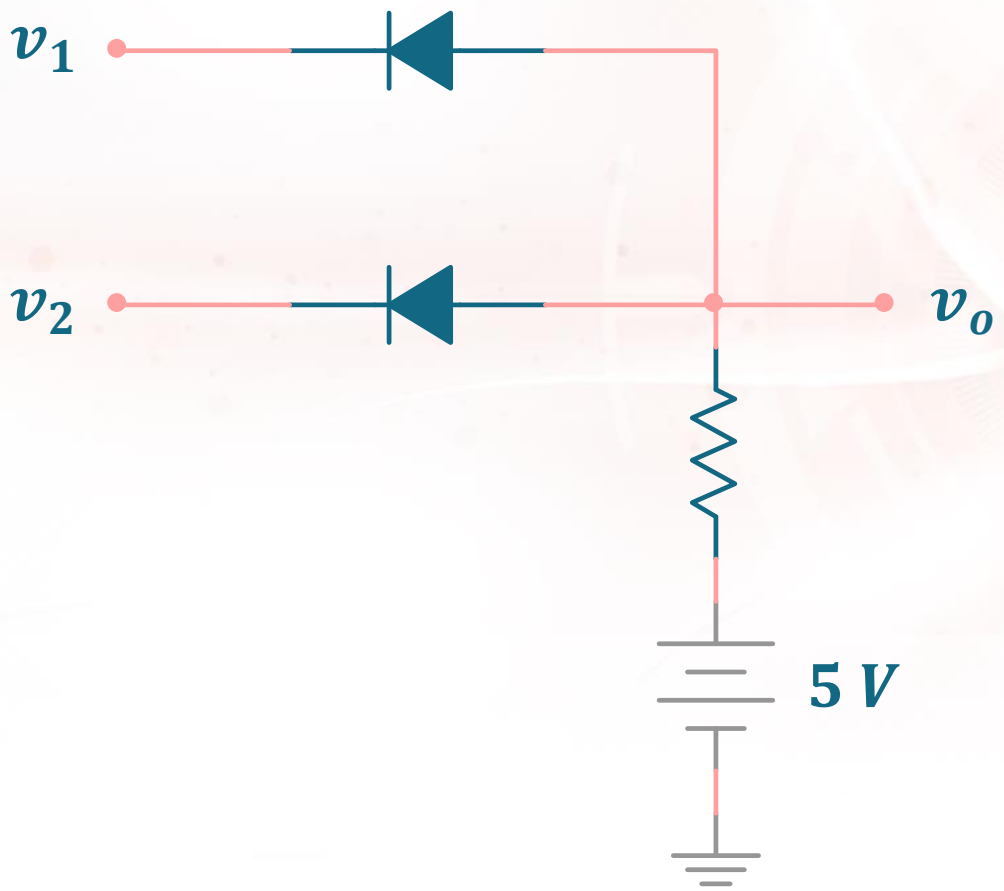
v_1	v_2	v_o
0	0	0
0	5	5
5	0	5
5	5	

POSITIVE LOGIC OR GATE



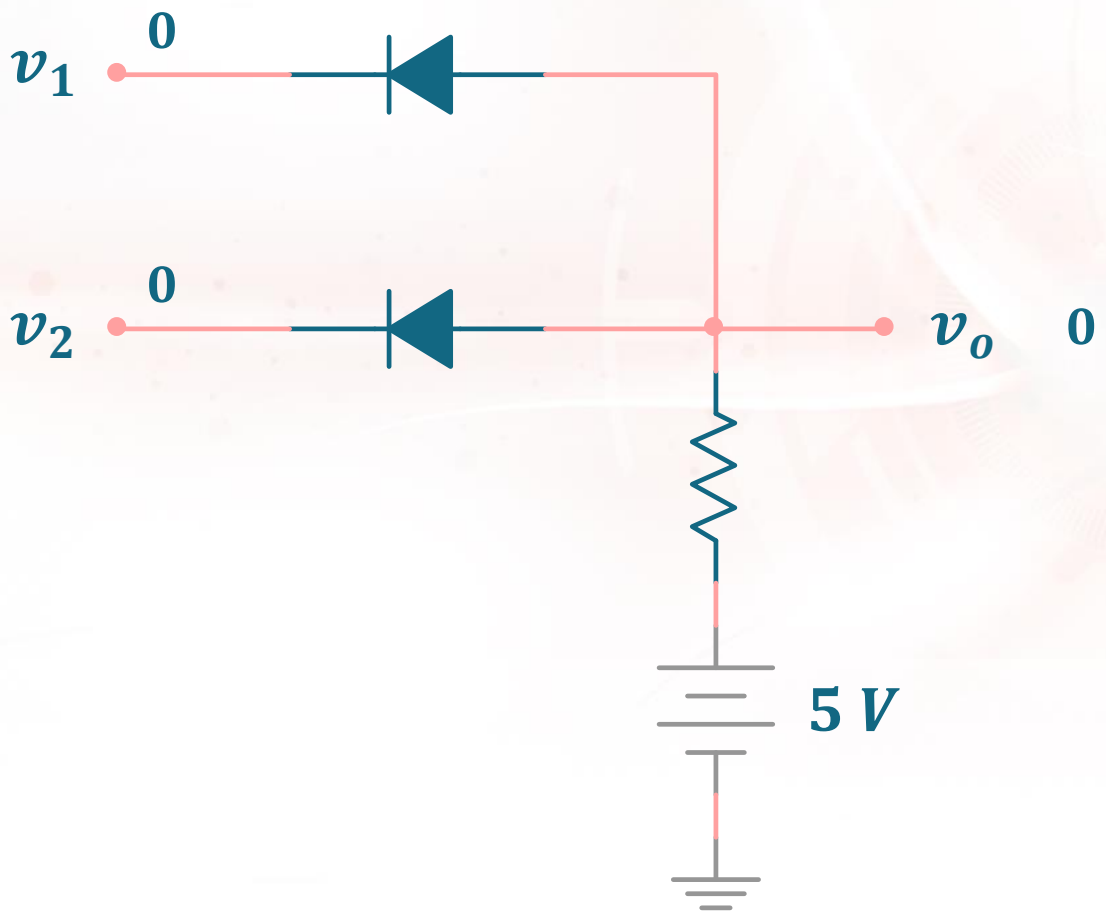
v_1	v_2	v_o
0	0	0
0	5	5
5	0	5
5	5	5

POSITIVE LOGIC AND GATE



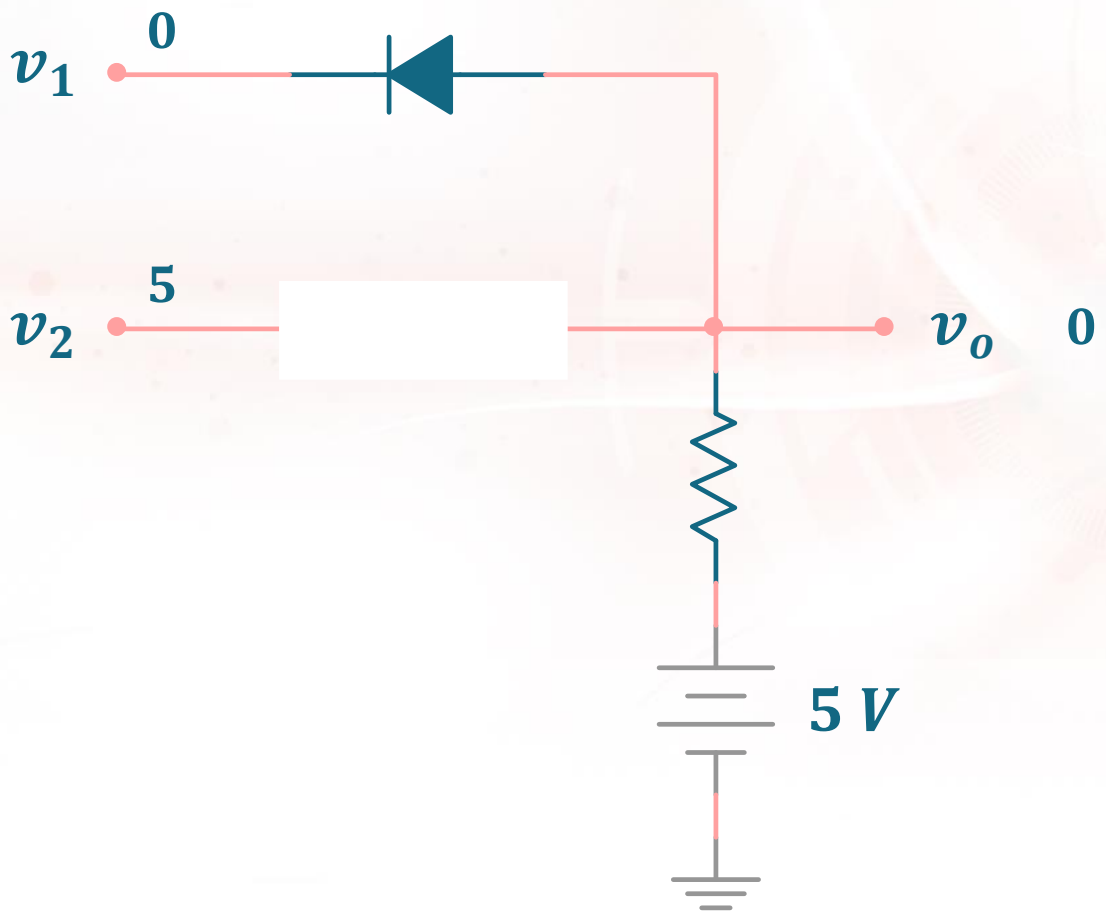
v_1	v_2	v_o
0	0	
0	5	
5	0	
5	5	

POSITIVE LOGIC AND GATE



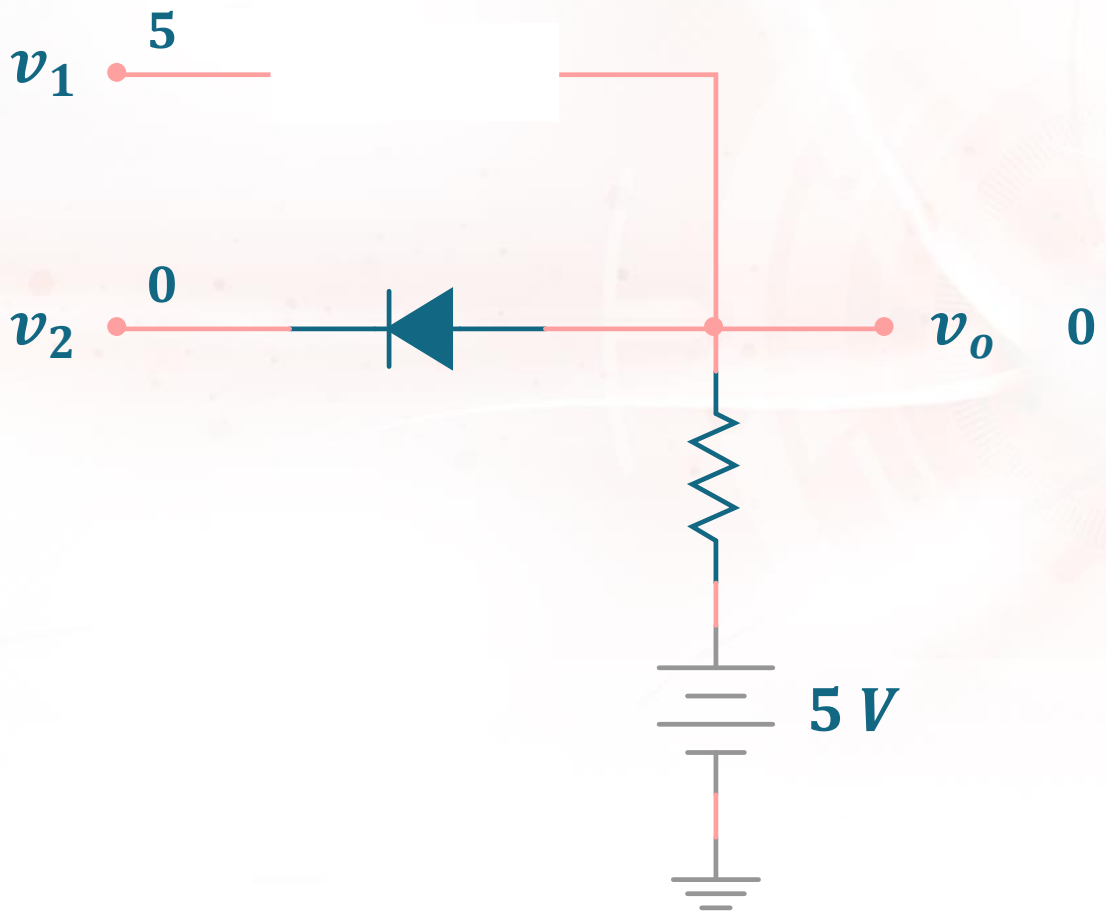
v_1	v_2	v_o
0	0	0
0	5	
5	0	
5	5	

POSITIVE LOGIC AND GATE



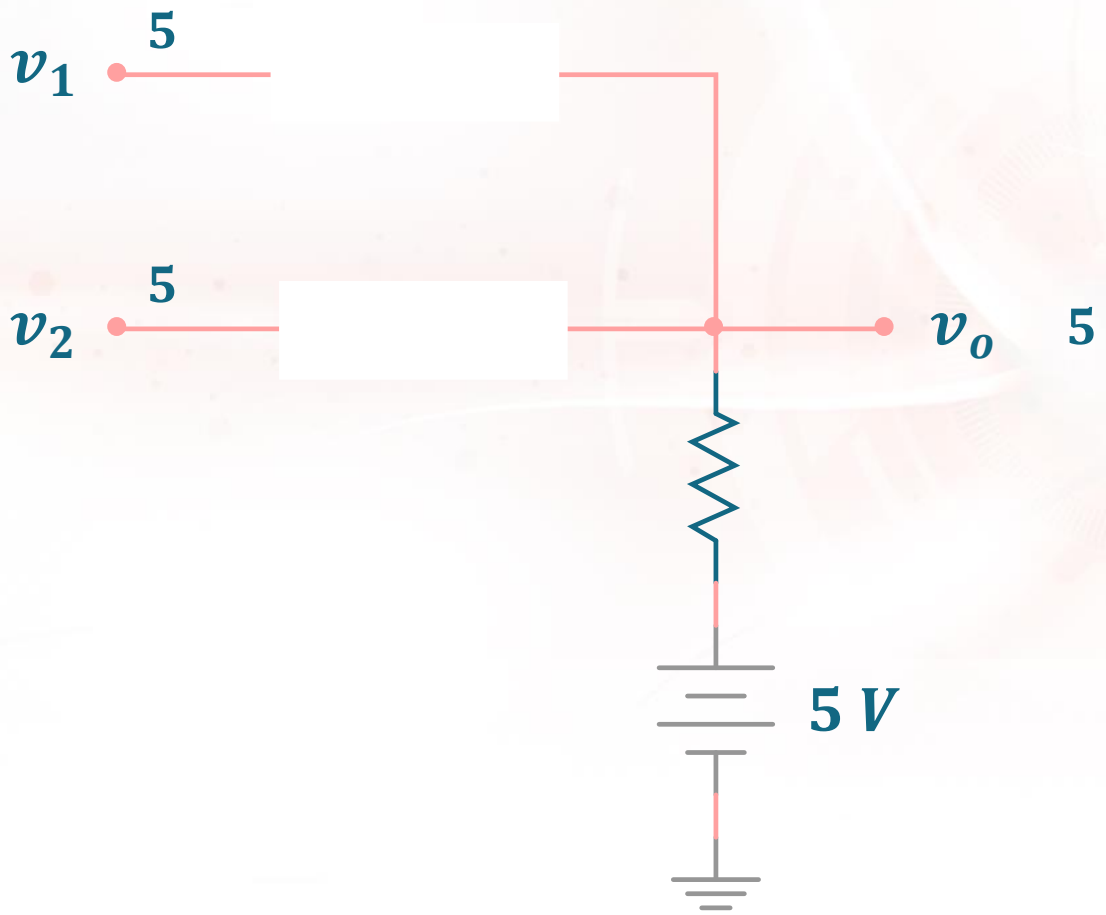
v_1	v_2	v_o
0	0	0
0	5	0
5	0	
5	5	

POSITIVE LOGIC AND GATE



v_1	v_2	v_o
0	0	0
0	5	0
5	0	0
5	5	

POSITIVE LOGIC AND GATE

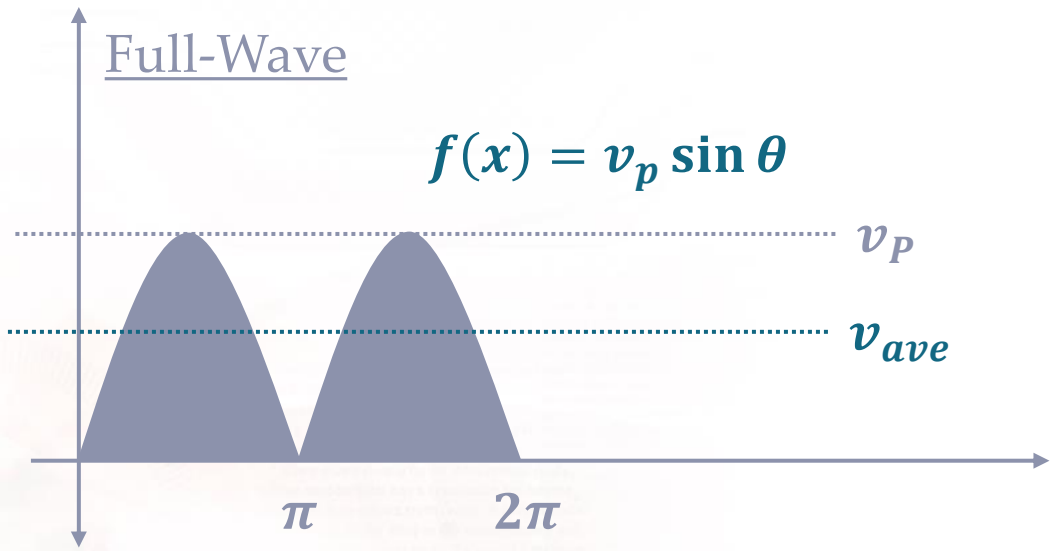


v_1	v_2	v_o
0	0	0
0	5	0
5	0	0
5	5	5

AC INPUT

DIODE CIRCUITS

AVERAGE VALUE OF $f(x)$



$$f(x)_{ave} = \frac{1}{b-a}\int_a^b f(x)dx$$

AVERAGE VALUE OF $f(x)$

$$f(x)_{ave} = \frac{1}{b-a}\int_a^b f(x)dx$$

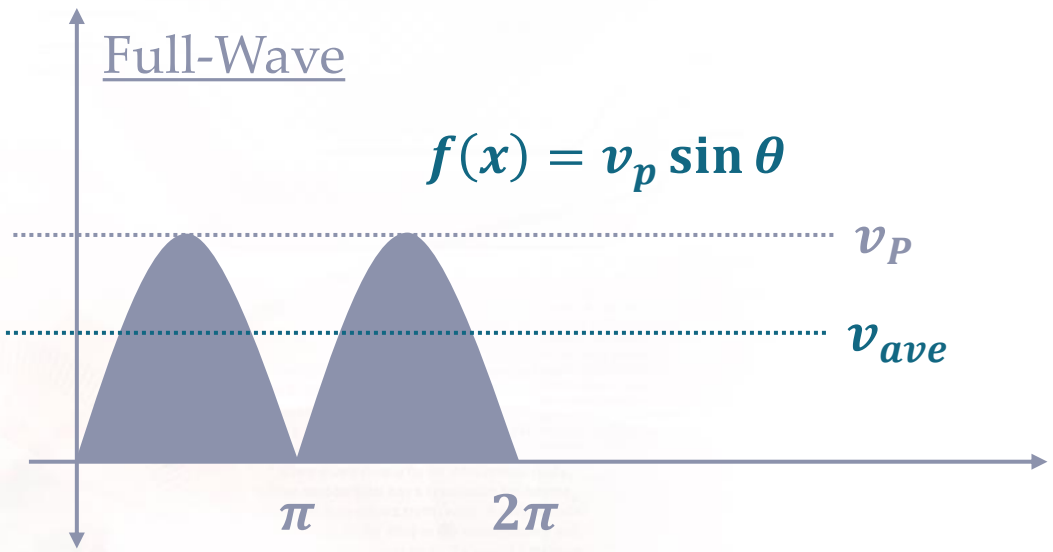
$$v_{ave} = \frac{1}{\pi - 0}\int_0^{\pi} v_p \sin\theta \, d\theta$$

$$v_{ave} = \frac{v_p}{\pi}(-\cos\theta)\Big|_0^{\pi}$$

$$v_{ave} = -\frac{v_p}{\pi}(\cos\pi - \cos 0)$$

$$v_{ave} = -\frac{v_p}{\pi}(-1-1)$$

$$v_{ave} = \frac{2v_p}{\pi}$$



AVERAGE VALUE OF $f(x)$

$$f(x)_{ave} = \frac{1}{b-a}\int_a^b f(x)dx$$

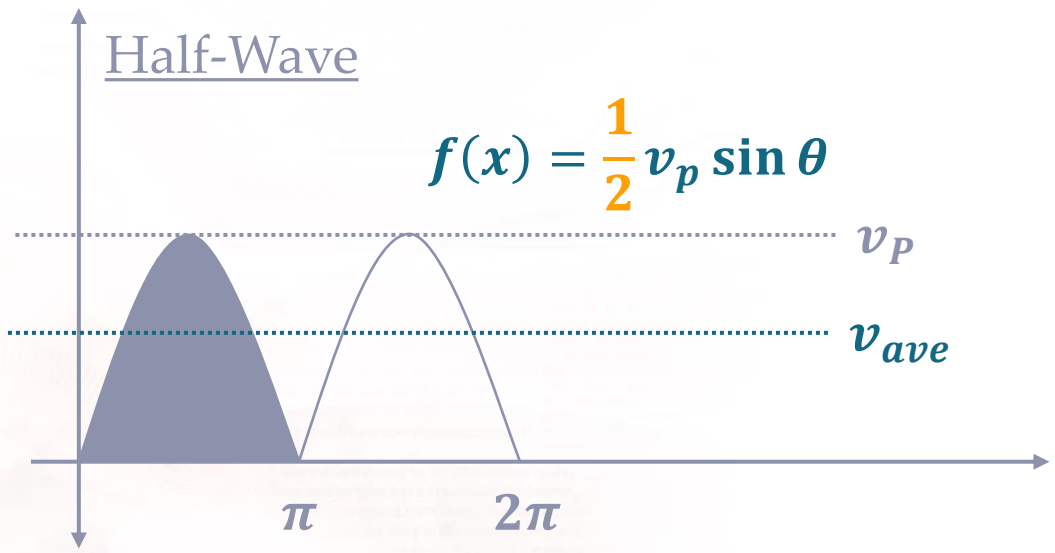
$$v_{ave} = \frac{1}{\pi - 0}\int_0^{\pi} \frac{1}{2} v_p \sin\theta \, d\theta$$

$$v_{ave} = \frac{v_p}{2\pi}(-\cos\theta)\Big|_0^{\pi}$$

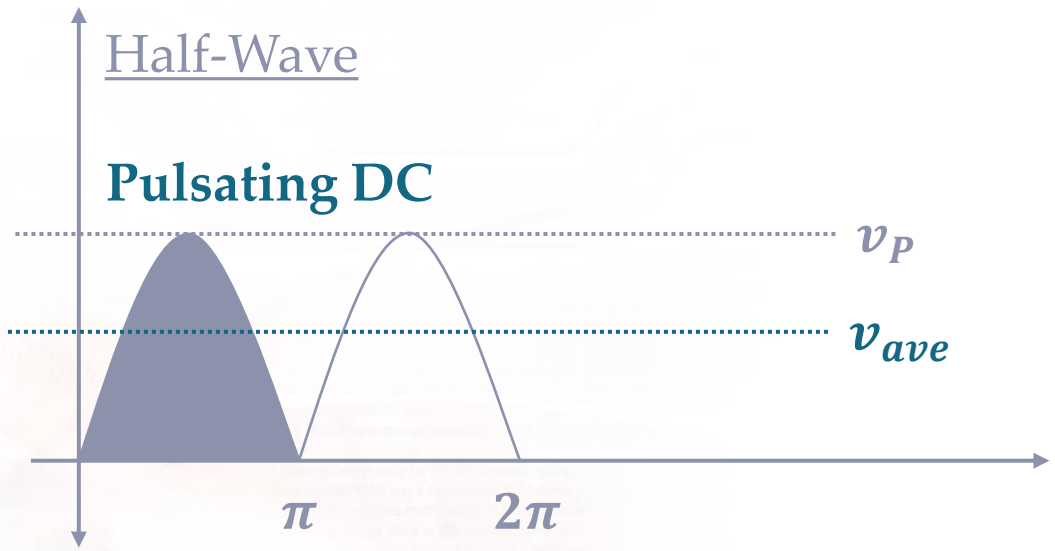
$$v_{ave} = -\frac{v_p}{2\pi}(\cos\pi - \cos 0)$$

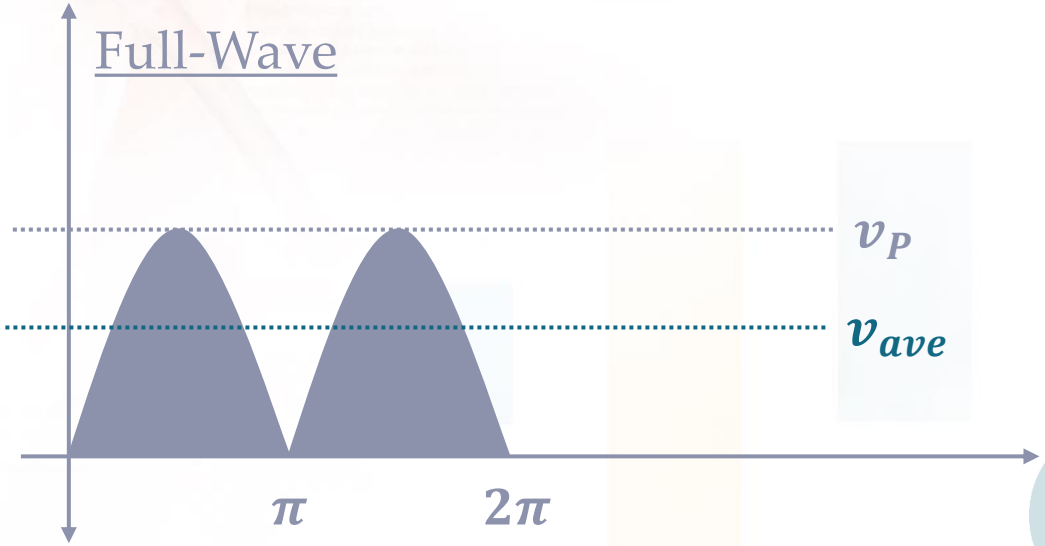
$$v_{ave} = -\frac{v_p}{2\pi}(-1-1)$$

$$v_{ave} = \frac{v_p}{\pi}$$

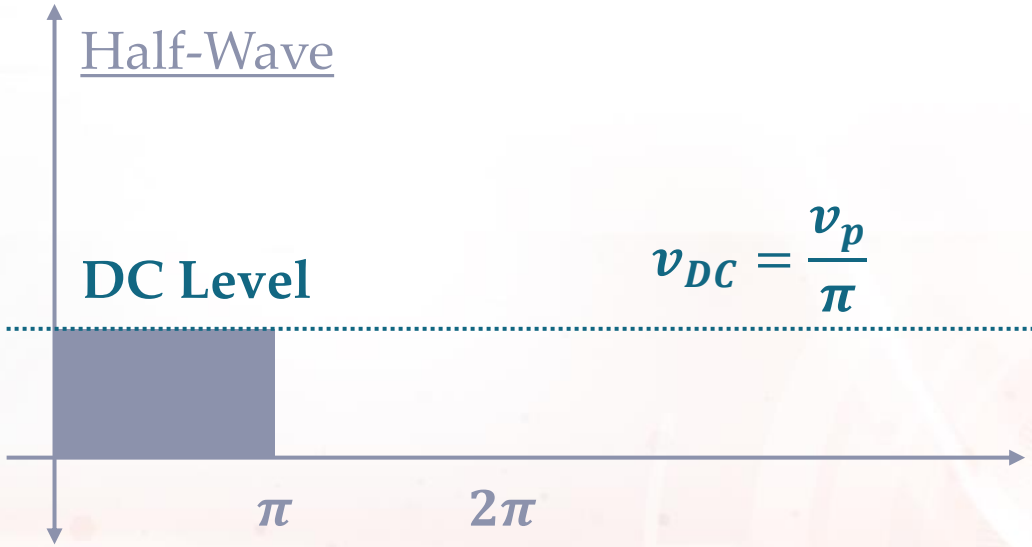


DC LEVEL

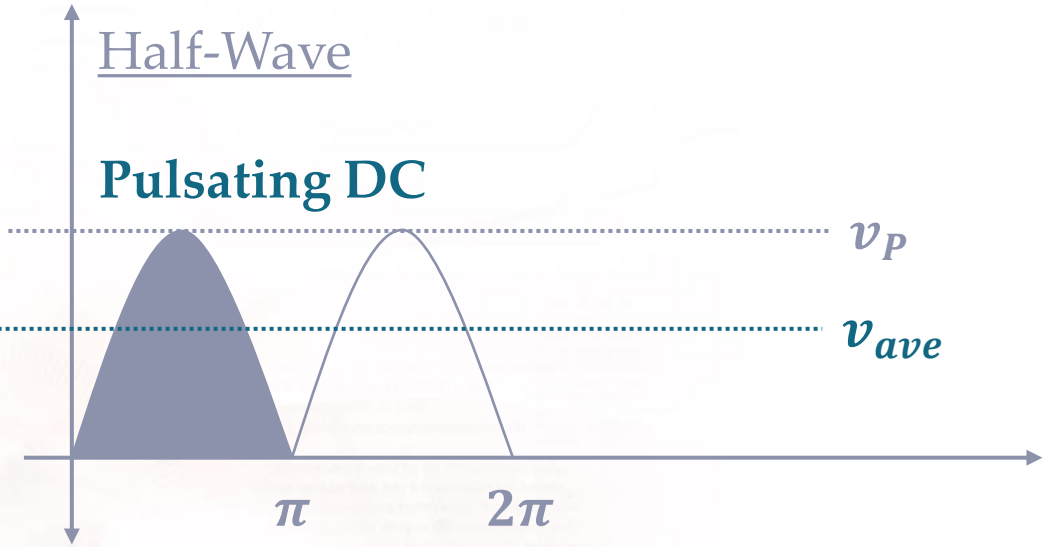


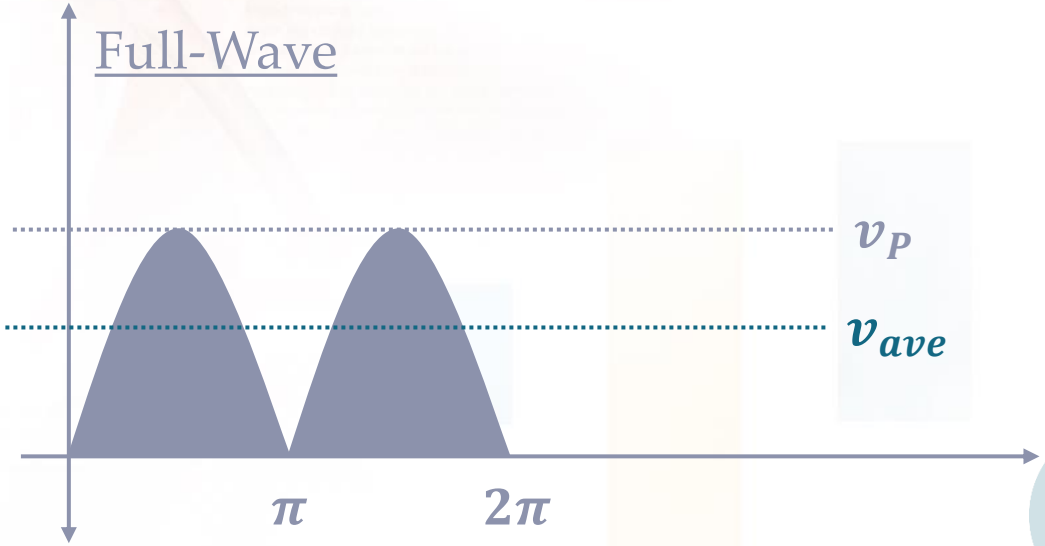


DC LEVEL

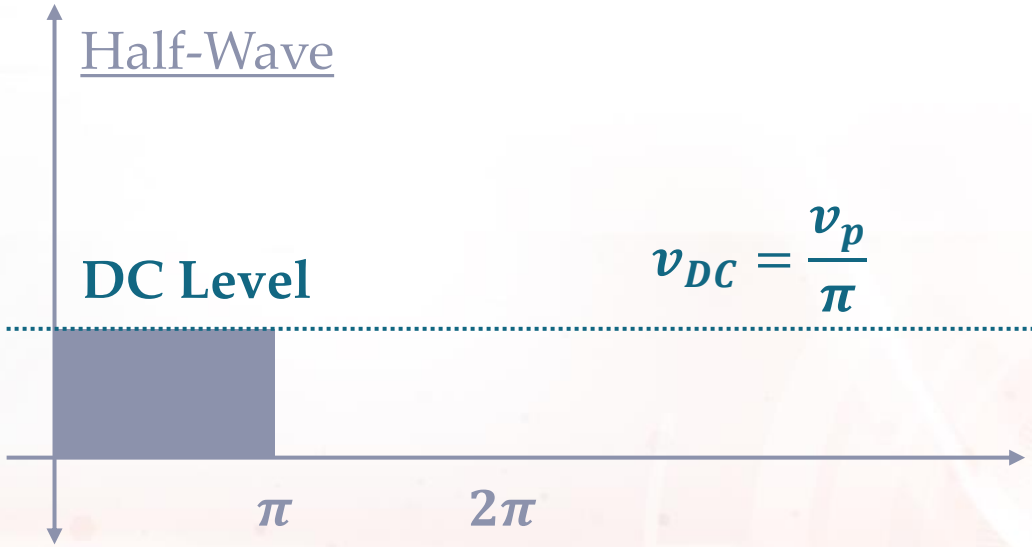


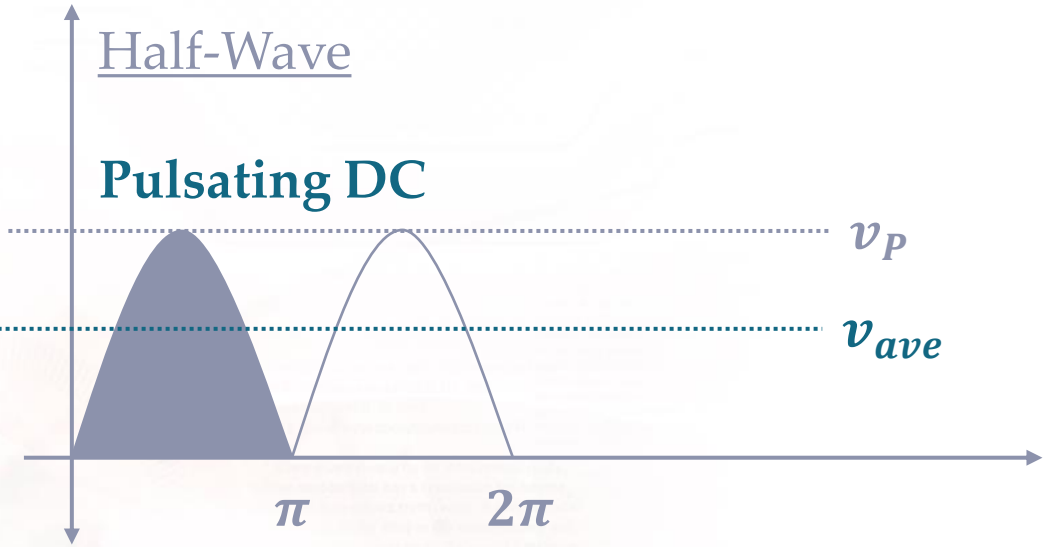
$$v_{DC} = \frac{v_p}{\pi}$$

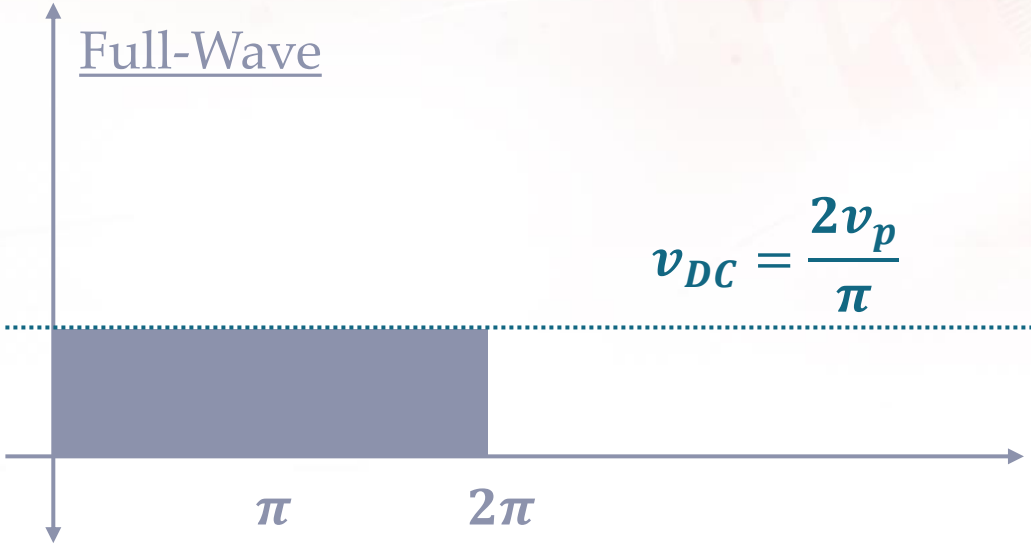


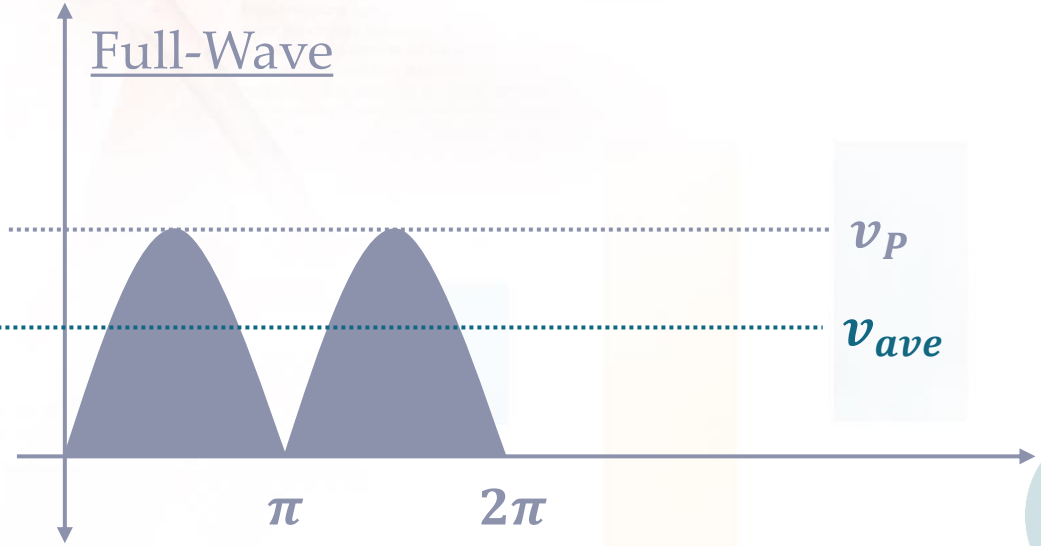


DC LEVEL

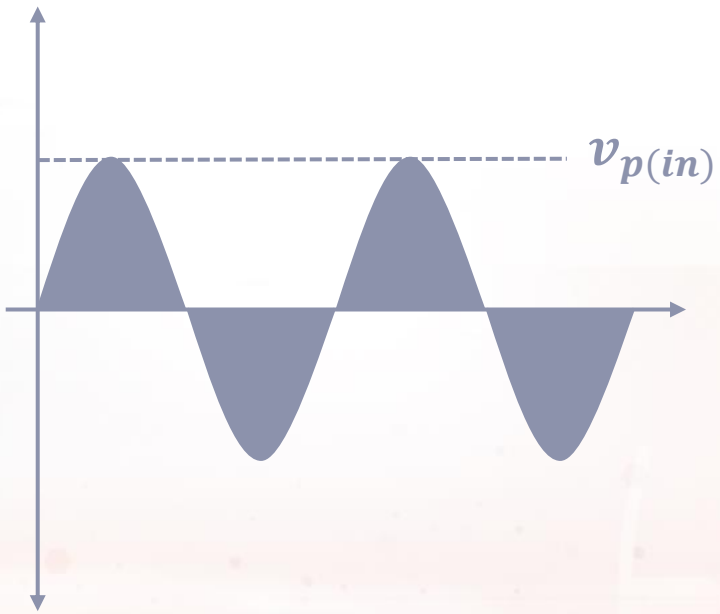

Half-Wave
DC Level
$v_{DC} = \frac{v_p}{\pi}$
π
2π

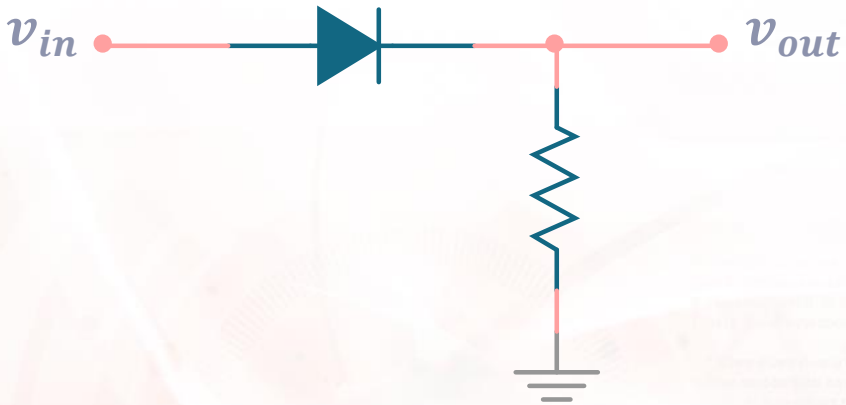

Half-Wave
Pulsating DC
v_P
v_{ave}
π
2π

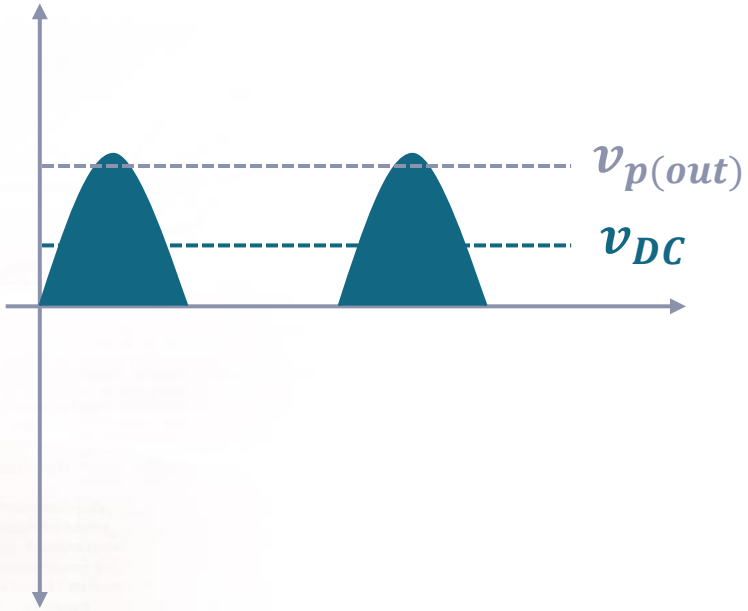

Full-Wave
$v_{DC} = \frac{2v_p}{\pi}$
π
2π


Full-Wave
v_P
v_{ave}
π
2π

HALF-WAVE RECTIFIER







DC Level,

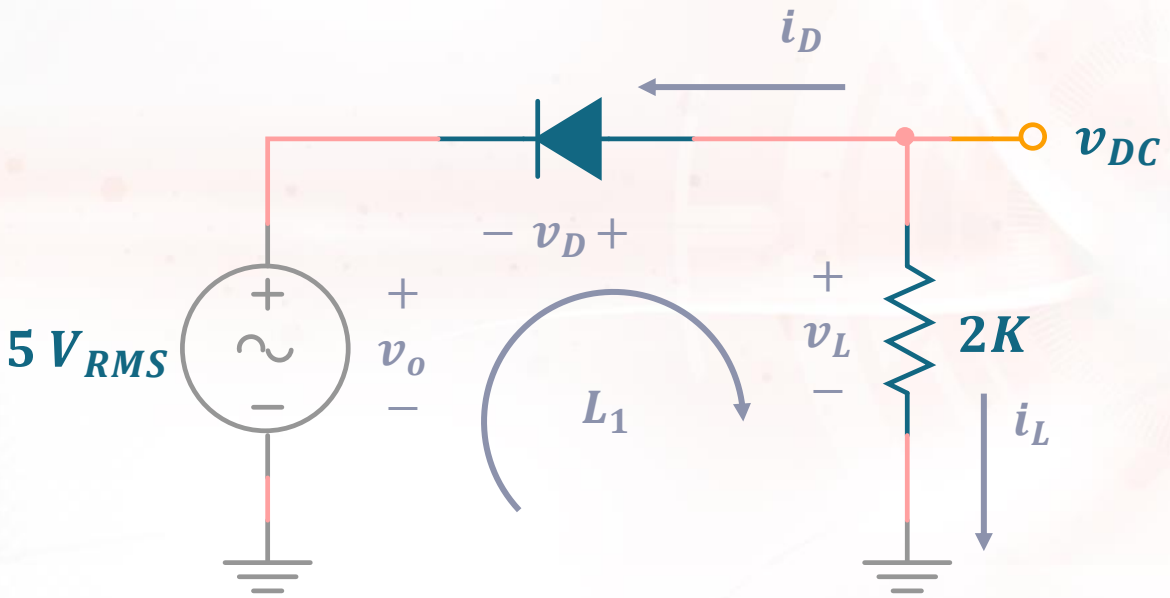
$$v_{DC} = \frac{v_{p(out)}}{\pi}$$

Output Frequency,

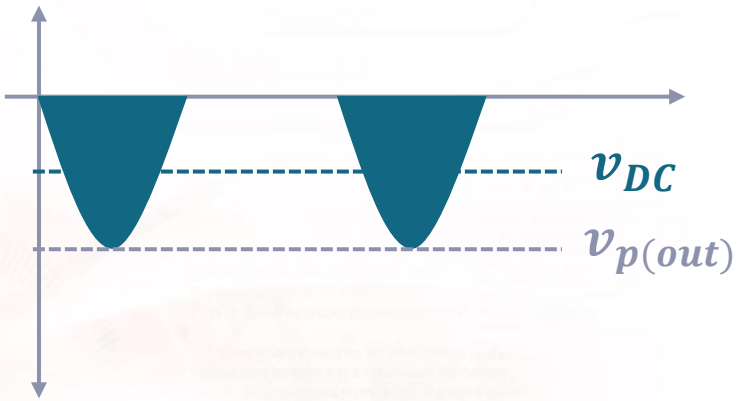
$$f_{out} = f_{in}$$

Sketch and determine the **dc level** of the output for the given network.

Sketch and determine the **dc level** of the output for the given network.



Sketch,



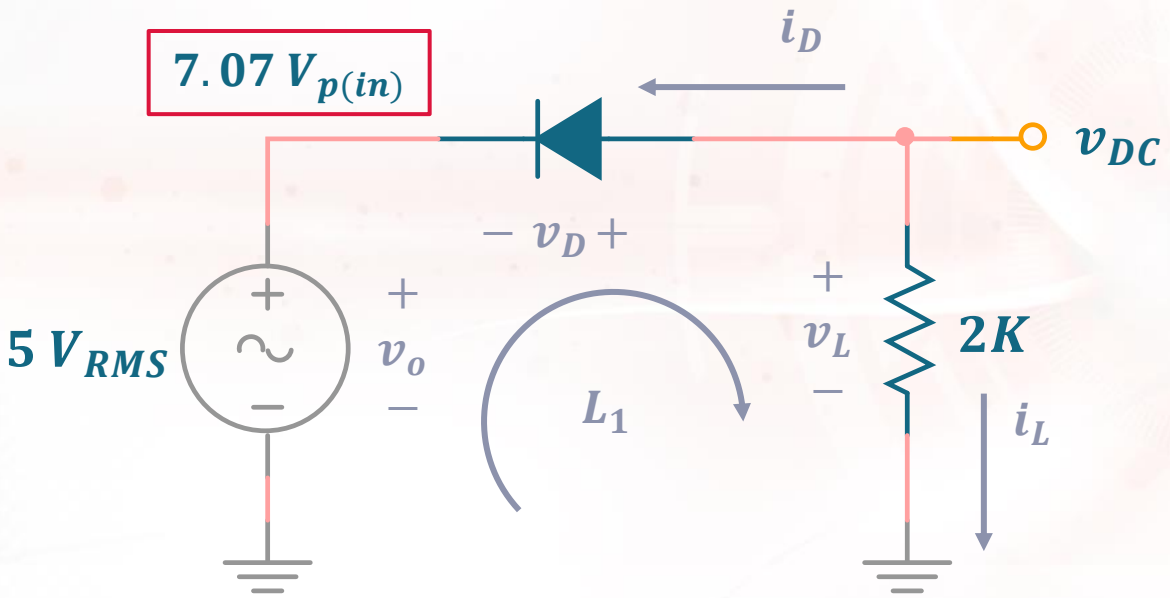
Convert RMS to peak,

$$v_{in} = v_{RMS}\sqrt{2}$$

$$v_{in} = 5\sqrt{2}$$

$$v_{in} = 7.07\ V_p$$

Sketch and determine the **dc level** of the output for the given network.



KVL @ L_1,

$$-v_o - v_D + v_L = 0$$

$$v_L = v_o + v_D$$

$$v_L = -7.07 + 0.7$$

$$v_{out} = -6.37\ V_p$$

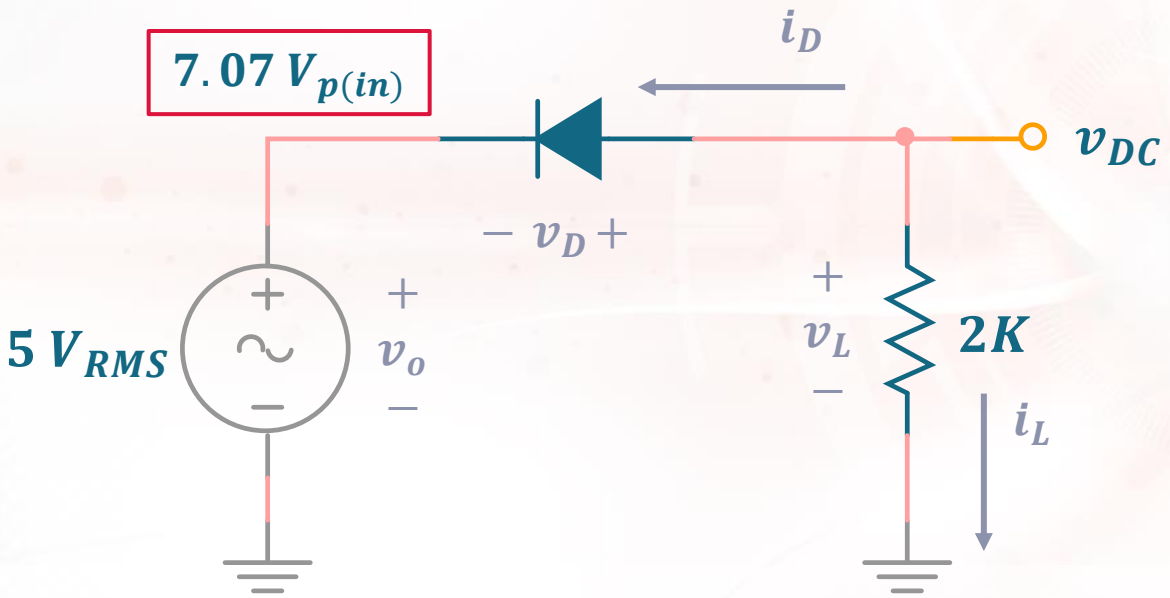
$$v_{DC} = \frac{v_{p(out)}}{\pi}$$

$$v_{DC} = \frac{-6.37}{\pi}$$

$$\boxed{v_{DC} = -2.03\ V}$$

ans

Sketch and determine the **dc level** of the output for the given network.



Node Analysis,

$$v_D = v_L - v_o$$

$$v_L = v_D + v_o$$

$$v_L = 0.7 + (-7.07)$$

$$v_{out} = -6.37\ V_p$$

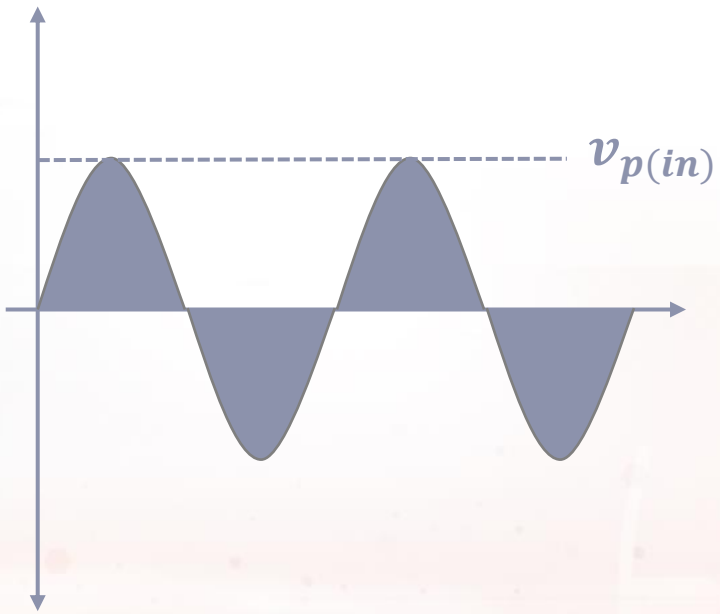
$$v_{DC} = \frac{v_{p(out)}}{\pi}$$

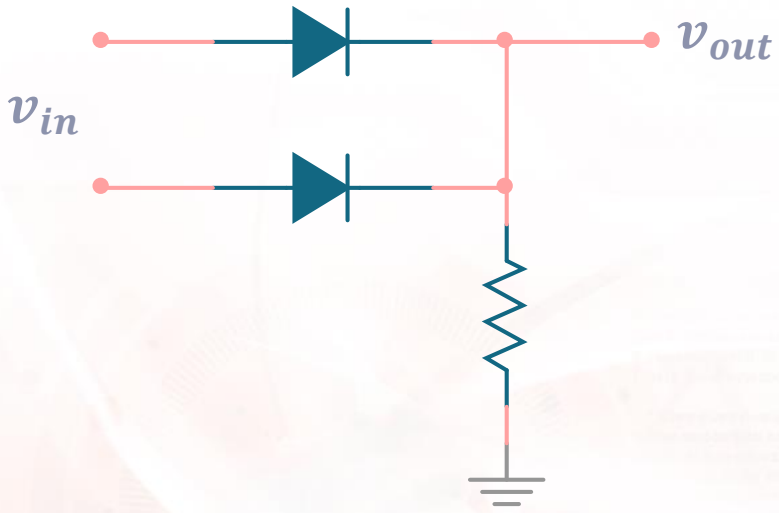
$$v_{DC} = \frac{-6.37}{\pi}$$

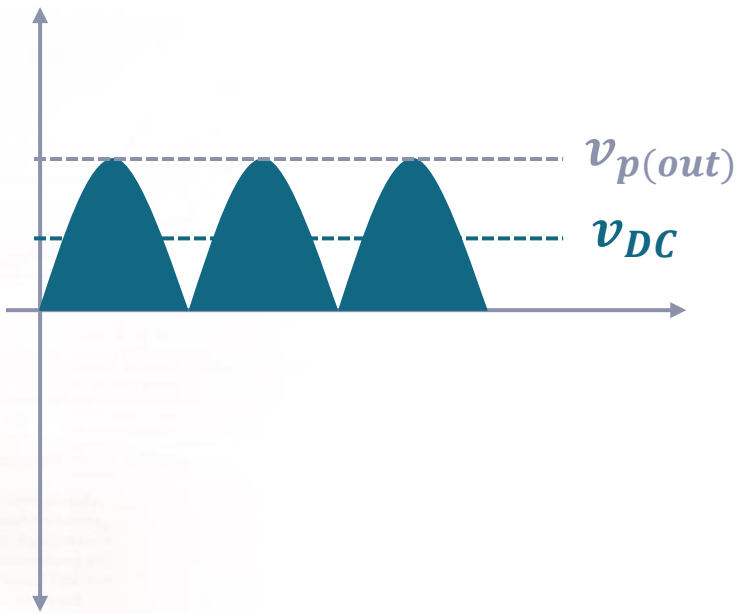
$$\boxed{v_{DC} = -2.03\ V}$$

ans

FULL WAVE CENTER-TAPPED RECTIFIER







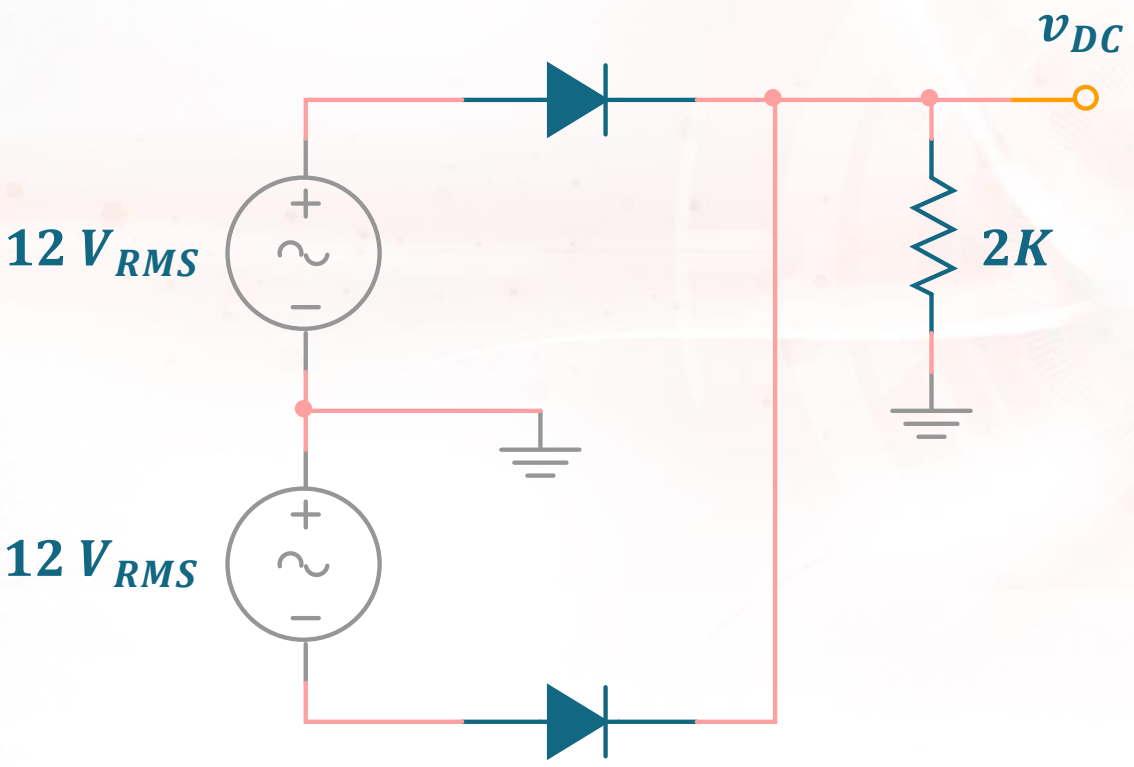
DC Level,

$$v_{DC} = \frac{2v_{p(out)}}{\pi}$$

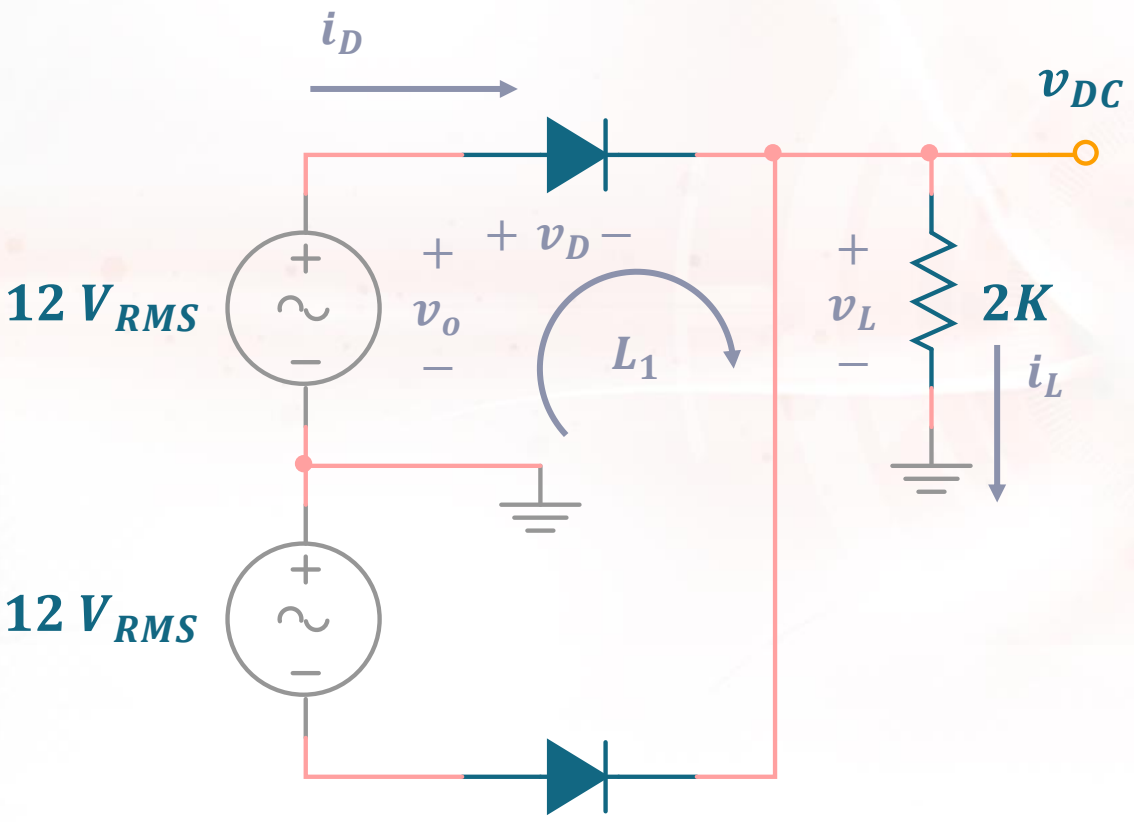
Output Frequency,

$$f_{out} = 2f_{in}$$

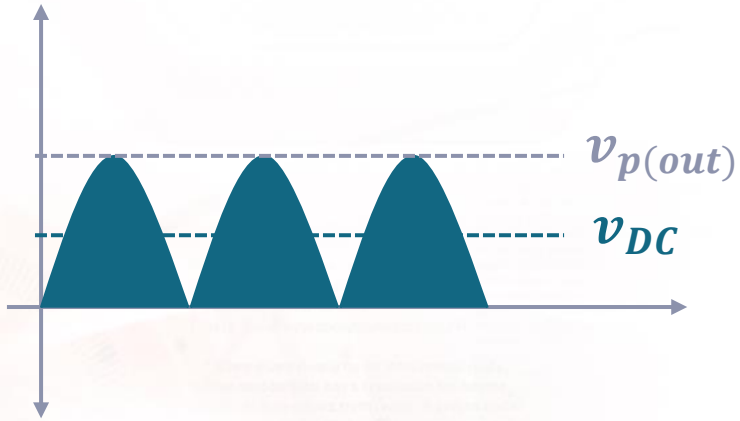
Sketch and determine the **dc level** of the output for the given network.



Sketch and determine the **dc level** of the output for the given network.



Sketch,



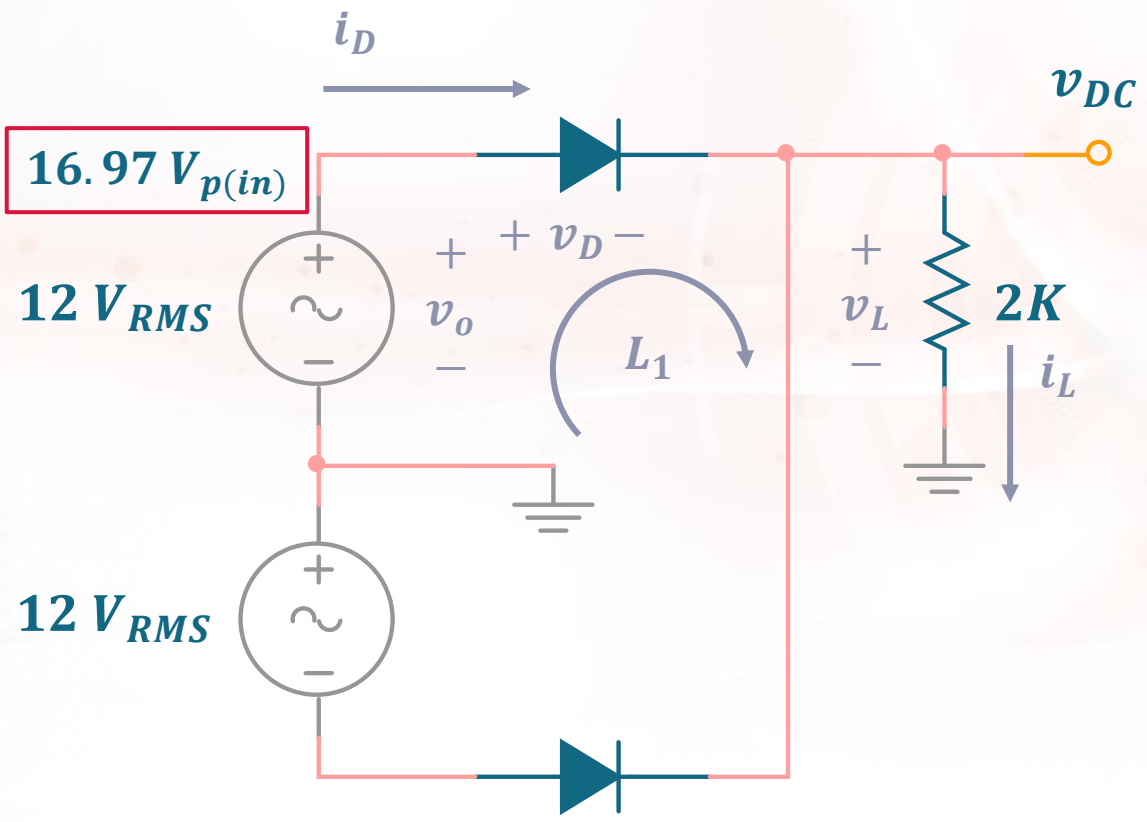
Convert RMS to peak,

$$v_{in} = v_{RMS}\sqrt{2}$$

$$v_{in} = 12\sqrt{2}$$

$$v_{in} = 16.97\ V_p$$

Sketch and determine the **dc level** of the output for the given network.



KCL @ L_1,

$$-v_o + v_D + v_L = 0$$

$$v_L = v_o - v_D$$

$$v_L = 16.97 - 0.7$$

$$v_{out} = 16.27\ v_P$$

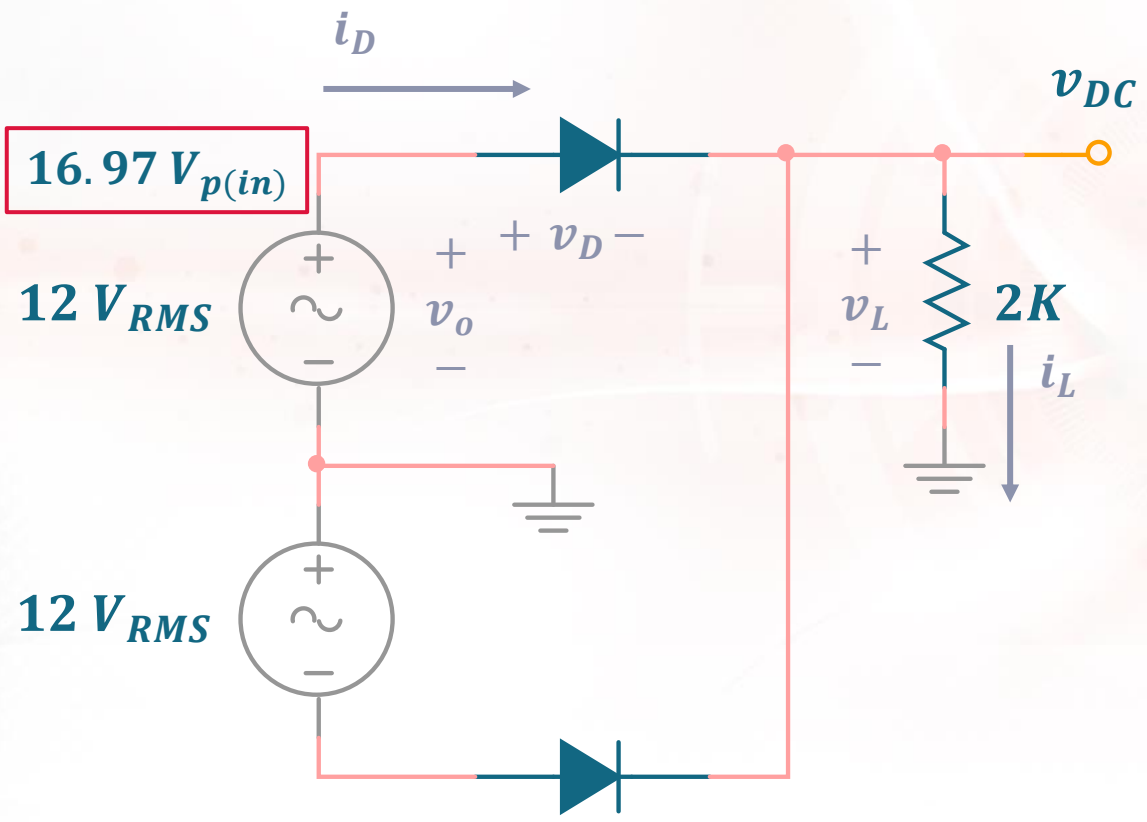
$$v_{DC} = \frac{2v_{p(out)}}{\pi}$$

$$v_{DC} = \frac{2(16.27)}{\pi}$$

$$\boxed{v_{DC} = 10.36\ V}$$

ans

Sketch and determine the **dc level** of the output for the given network.



Node Analysis,

$$v_D = v_o - v_L$$

$$v_L = v_o - v_D$$

$$v_L = 16.97 - 0.7$$

$$v_{out} = 16.27\ v_P$$

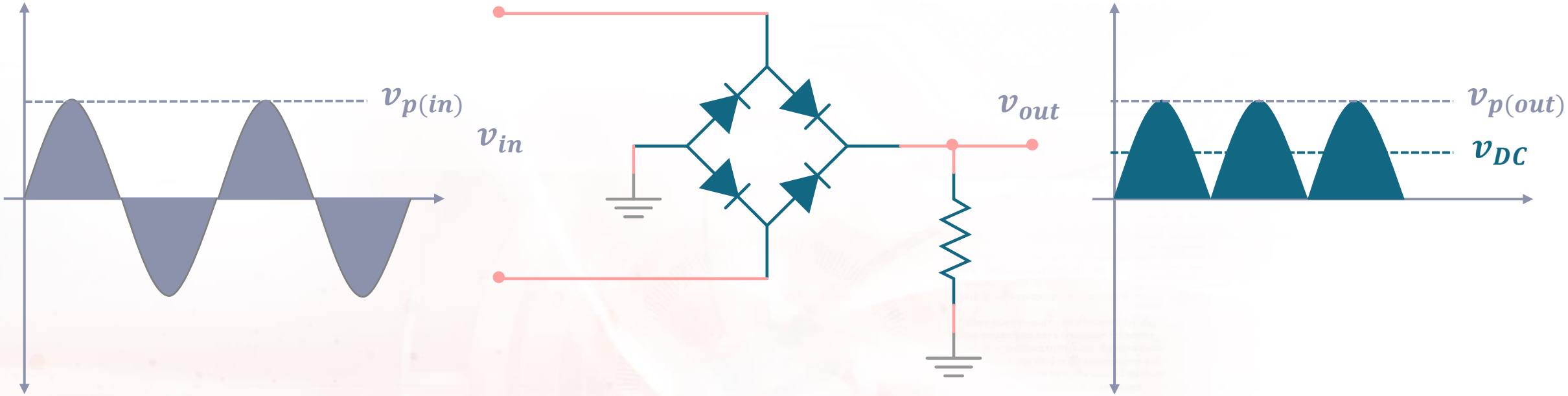
$$v_{DC} = \frac{2v_{p(out)}}{\pi}$$

$$v_{DC} = \frac{2(16.27)}{\pi}$$

$$\boxed{v_{DC} = 10.36\ V}$$

ans

FULL WAVE BRIDGE-TYPE RECTIFIER



DC Level,

$$v_{DC} = \frac{2v_{p(out)}}{\pi}$$

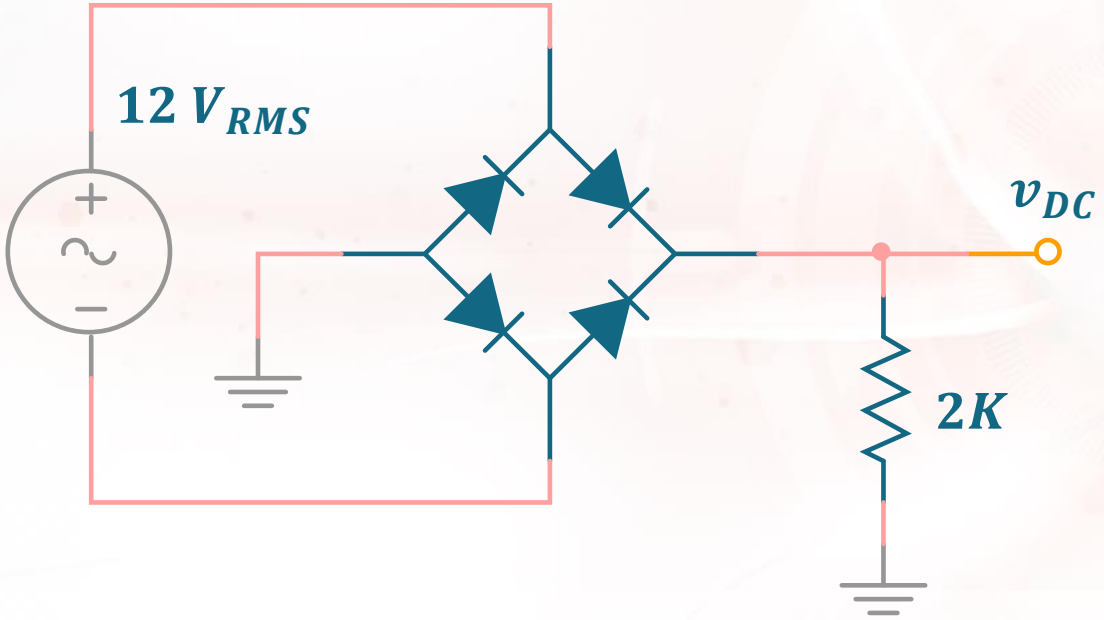
note,

two diodes are ON every half-cycle

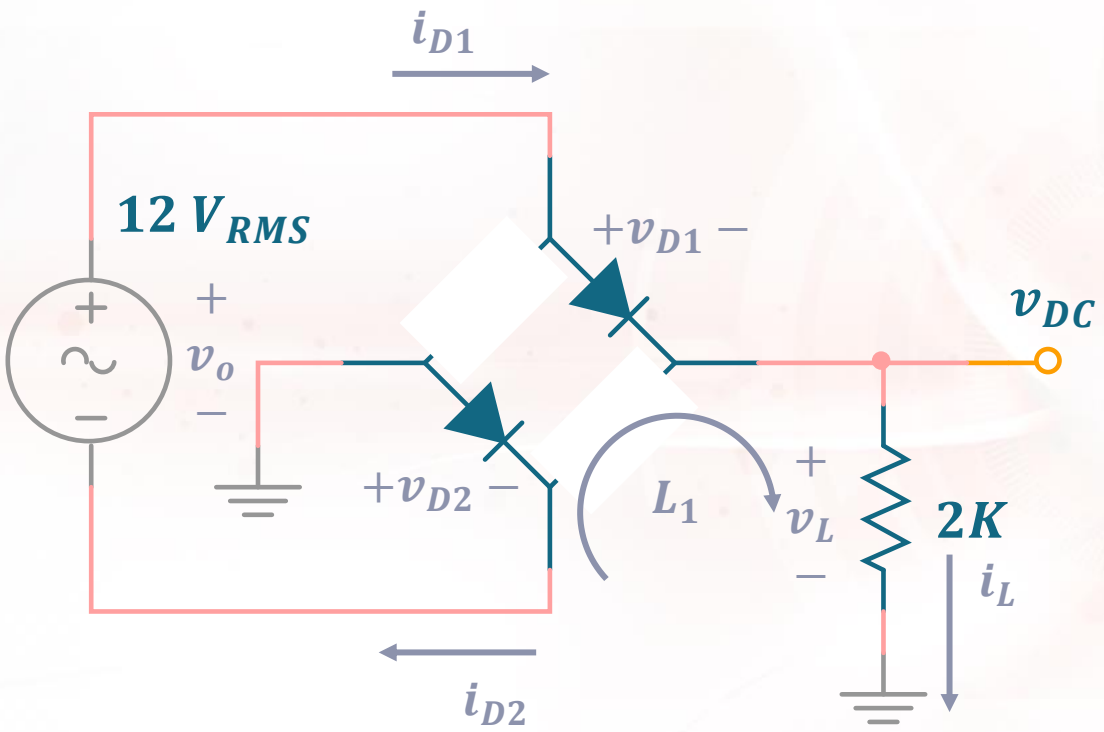
Output Frequency,

$$f_{out} = 2f_{in}$$

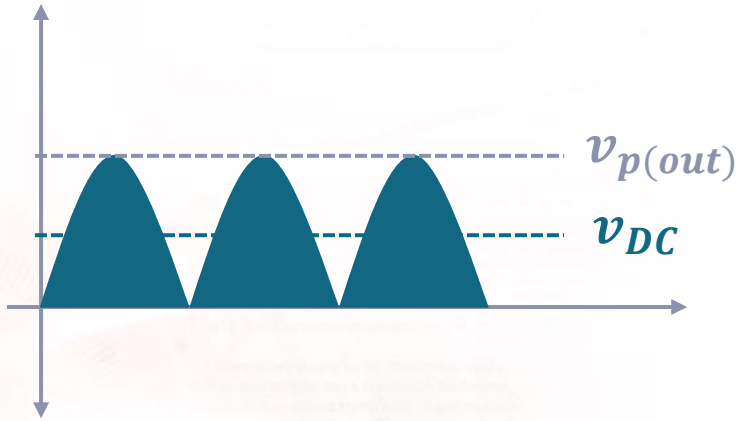
Sketch and determine the **dc level** of the output for the given network.



Sketch and determine the **dc level** of the output for the given network.



Sketch,



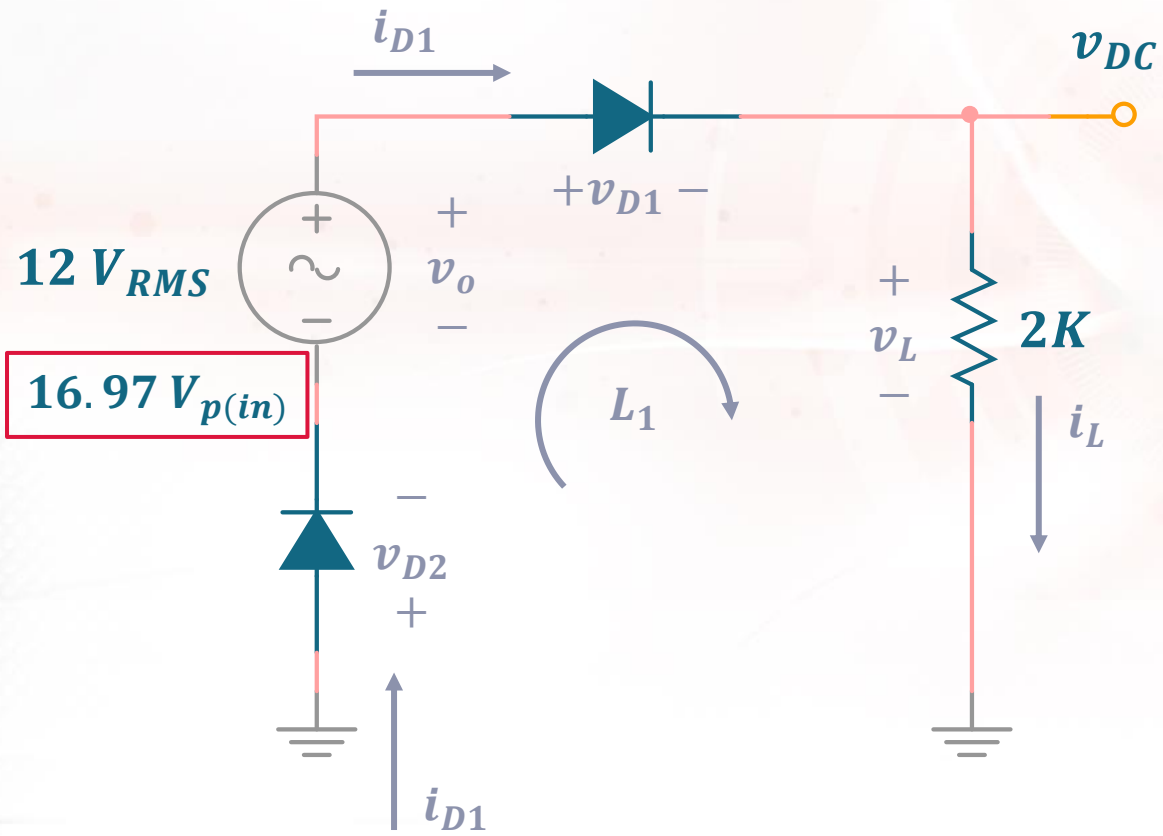
Convert RMS to peak,

$$v_{in} = v_{RMS}\sqrt{2}$$

$$v_{in} = 12\sqrt{2}$$

$$v_{in} = 16.97\ V_p$$

Sketch and determine the **dc level** of the output for the given network.



KVL @ L_1,

$$v_{D2} - v_o + v_{D1} + v_L = 0$$

$$v_L = v_o - v_{D1} - v_{D2}$$

$$v_L = 16.97 - 0.7 - 0.7$$

$$v_{out} = 15.57\ v_p$$

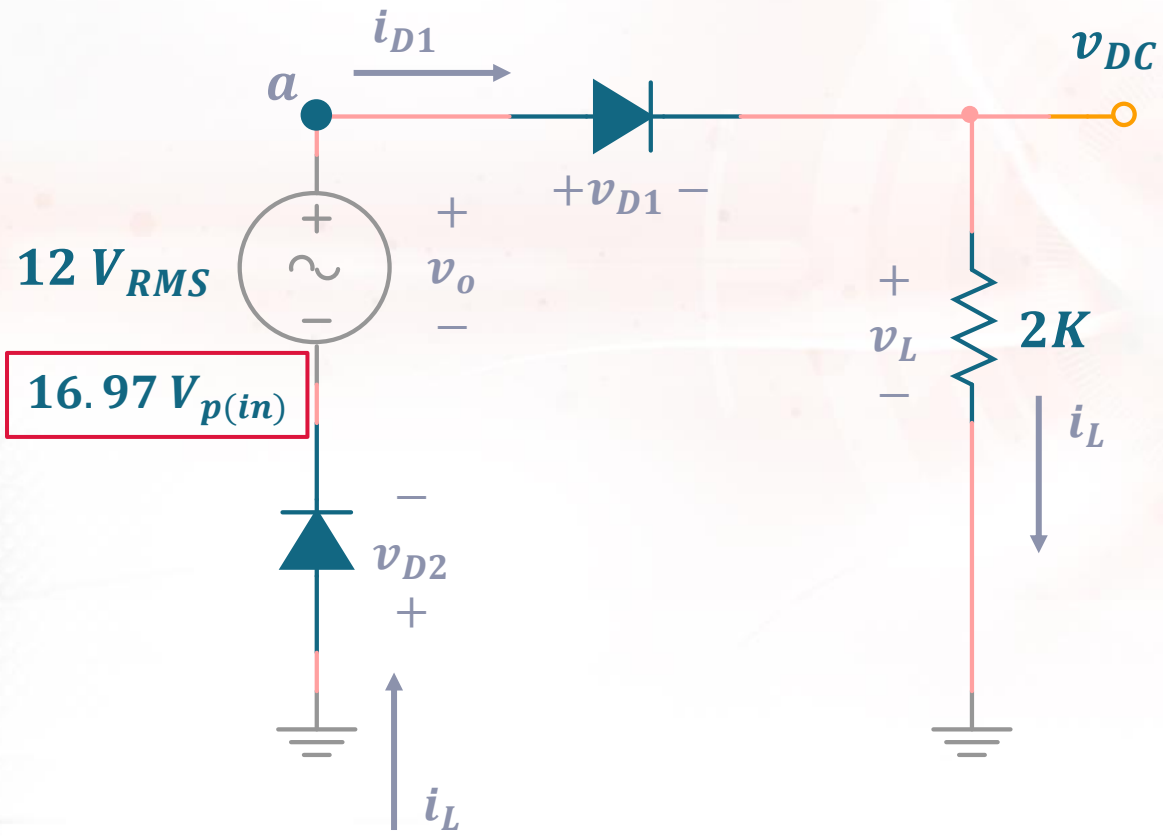
$$v_{DC} = \frac{2v_{p(out)}}{\pi}$$

$$v_{DC} = \frac{2(15.57)}{\pi}$$

$$\boxed{v_{DC} = 9.91\ V}$$

ans

Sketch and determine the **dc level** of the output for the given network.



Node Analysis,

$$v_{D1} = v_a - v_L \qquad v_a = v_o - v_{D2}$$

$$v_{D1} = v_o - v_{D2} - v_L$$

$$v_L = v_o - v_{D1} - v_{D2}$$

$$v_L = 16.97 - 0.7 - 0.7$$

$$v_{out} = 15.57\ v_p$$

$$v_{DC} = \frac{2v_{p(out)}}{\pi}$$

$$v_{DC} = \frac{2(15.57)}{\pi}$$

$$\boxed{v_{DC} = 9.91\ V}$$

ans

LABORATORY